8

Chaining

This chapter presents some of the central concepts and methods to bound random processes. Chaining is a powerful and general technique to prove uniform bounds on a random process $(X_t)_{t \in T}$. We present a basic version of chaining method in Section 8.1. There we prove Dudley's bound on random processes in terms of covering numbers of T. In Section 8.2, we give applications of Dudley's inequality to Monte-Carlo integration and a uniform law of large numbers.

In Section 8.3 we show how to find bounds for random processes in terms of the VC dimension of T. Unlike covering numbers, VC dimension is a combinatorial rather than geometric quantity. It plays important role in problems of statistical learning theory, which we discuss in Section 8.4.

As we will see in Section 8.1.2), the bounds on empirical processes in terms of covering numbers – Sudakov's inequality from Section 7.4 and Dudley's inequality – are sharp up to a logarithmic factor. The logarithmic gap is insignificant in many applications, but it can not be removed in general. A sharper bound on random processes, without any logarithmic gap, can be given in terms of the so-called M. Talagrand's functional $\gamma_2(T)$, which captures the geometry of T better than the covering numbers. We prove a sharp upper bound is Section 8.5 by a refined chaining argument, often called "generic chaining".

A matching lower bound due to M. Talagrand is more difficult to obtain; we will state it without proof in Section 8.6. The resulting sharp, two-sided bound on random processes is known as the *majorizing measure theorem* (Theorem 8.6.1). A very useful consequence of this result is *Talagrand's comparison inequality* (Corollary 8.6.2), which generalizes Sudakov-Fernique's inequality for all sub-gaussian random processes.

Talagrand's comparison inequality has many applications. One of them, *Chevet's inequality*, will be discussed in Section 8.7; others will appear later.

8.1 Dudley's inequality

Sudakov's minoration inequality that we studied in Section 7.4 gives a *lower bound* on the magnitude

$$\mathbb{E} \sup_{t \in T} X_t$$

of a Gaussian random process $(X_t)_{t\in T}$ in terms of the metric entropy of T. In this section, we obtain a similar *upper bound*.

This time, we are able to work not just with Gaussian processes but with more general processes with sub-gaussian increments.

Definition 8.1.1 (Sub-gaussian increments)**.** Consider a random process $(X_t)_{t\in T}$ on a metric space (T,d). We say that the process has *sub-gaussian increments* if there exists $K \geq 0$ such that

$$\|X_t - X_s\|_{\psi_2} \leq K d(t,s) \quad \text{for all } t,s \in T. \tag{8.1}$$

Example 8.1.2. Let $(X_t)_{t\in T}$ be a Gaussian process on an abstract set T. Define a metric on T by

$$d(t,s) := \|X_t - X_s\|_{L^2}, \quad t,s \in T.$$

Then $(X_t)_{t\in T}$ is obviously a process with sub-gaussian increments, and K is an absolute constant.

We now state Dudley's inequality, which gives a bound on a general sub-gaussian random process $(X_t)_{t\in T}$ in terms of the metric entropy $\log \mathcal{N}(T,d,\varepsilon)$ of T.

Theorem 8.1.3 (Dudley's integral inequality)**.** *Let $(X_t)_{t\in T}$ be a mean zero random process on a metric space (T,d) with sub-gaussian increments as in (8.1). Then*

$$\mathbb{E}\sup_{t\in T} X_t \leq CK \int_0^\infty \sqrt{\log \mathcal{N}(T,d,\varepsilon)}\, d\varepsilon.$$

Before we prove Dudley's inequality, it is helpful to compare it with Sudakov's inequality (Theorem 7.4.1), which for Gaussian processes states that

$$\mathbb{E}\sup_{t\in T} X_t \geq c \sup_{\varepsilon>0} \varepsilon \sqrt{\log \mathcal{N}(T,d,\varepsilon)}.$$

Figure 8.1 illustrates Dudley's and Sudakov's bounds. There is an obvious gap between these two bounds. It can not be closed in terms of the entropy numbers alone; we will explore this point later.

The right hand side of Dudley's inequality might suggest to us that $\mathbb{E}\sup_{t\in T} X_t$ is a *multi-scale* quantity, in that we have to examine T at all possible scales ε in order to bound the process. This is indeed so, and our proof will indeed be multi-scale. We now state and prove a discrete version of Dudley's inequality, where the integral over all positive ε is replaced by a sum over dyadic values $\varepsilon = 2^{-k}$, which somewhat resembles a Riemann sum. Later we will quickly pass to the original form of Dudley's inequality.

Theorem 8.1.4 (Discrete Dudley's inequality)**.** *Let $(X_t)_{t\in T}$ be a mean zero random process on a metric space (T,d) with sub-gaussian increments as in (8.1). Then*

$$\mathbb{E}\sup_{t\in T} X_t \leq CK \sum_{k\in\mathbb{Z}} 2^{-k} \sqrt{\log \mathcal{N}(T,d,2^{-k})}. \tag{8.2}$$

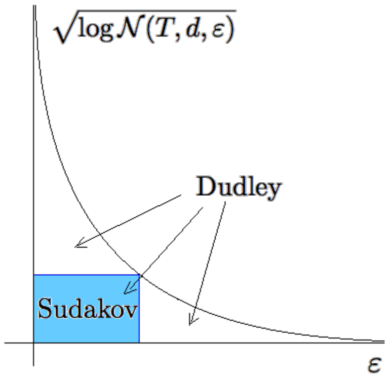


Figure 8.1 Dudley's inequality bounds $\mathbb{E}\sup_{t\in T} X_t$ by the area under the curve. Sudakov's inequality bounds it below by the largest area of a rectangle under the curve, up to constants.

Our proof of this theorem will be based on the important technique of *chaining*, which can be useful in many other problems. Chaining is a *multi-scale* version of the ε-net argument that we used successfully in the past, for example in the proofs of Theorems 4.4.5 and 7.7.1.

In the familiar, single-scale ε-net argument, we discretize T by choosing an ε-net $\mathcal{N}$ of T. Then every point $t \in T$ can be approximated by a closest point from the net $\pi(t) \in \mathcal{N}$ with accuracy ε, so that $d(t, \pi(t)) \le \varepsilon$. The increment condition (8.1) yields

$$\|X_t - X_{\pi(t)}\|_{\psi_2} \le K\varepsilon. \tag{8.3}$$

This gives

$$\mathbb{E}\sup_{t\in T} X_t \le \mathbb{E}\sup_{t\in T} X_{\pi(t)} + \mathbb{E}\sup_{t\in T}(X_t - X_{\pi(t)}).$$

The first term can be controlled by a union bound over $|\mathcal{N}| = \mathcal{N}(T,d,\varepsilon)$ points $\pi(t)$.

To bound the second term, we would like to use (8.3). But it only holds for fixed $t \in T$, and it is not clear how to control the supremum over $t \in T$. To overcome this difficulty, we do not stop here but continue to run the ε-net argument further, building progressively finer approximations $\pi_1(t), \pi_2(t), \ldots$ to t with finer nets. Let us now develop formally this technique of chaining.

Proof of Theorem 8.1.4. **Step 1: Chaining set-up.** Without loss of generality, we may assume that $K = 1$ and that T is finite. (Why?) Let us set the dyadic scale

$$\varepsilon_k = 2^{-k}, \quad k \in \mathbb{Z} \tag{8.4}$$

and choose ε_k-nets T_k of T so that

$$|T_k| = \mathcal{N}(T, d, \varepsilon_k). \tag{8.5}$$

Only a part of the dyadic scale will be needed. Indeed, since T is finite, there exists a small enough number $\kappa \in \mathbb{Z}$ (defining the coarsest net) and a large enough number $K \in \mathbb{Z}$ (defining the finest net), such that

$$T_\kappa = \{t_0\} \text{ for some } t_0 \in T, \quad T_K = T. \tag{8.6}$$

For a point $t \in T$, let $\pi_k(t)$ denote a closest point in T_k, so we have

$$d(t, \pi_k(t)) \le \varepsilon_k. \tag{8.7}$$

Since $\mathbb{E}\, X_{t_0} = 0$, we have

$$\mathbb{E} \sup_{t \in T} X_t = \mathbb{E} \sup_{t \in T} (X_t - X_{t_0}).$$

We can express $X_t - X_{t_0}$ as a telescoping sum; think about walking from t_0 to t along a chain of points $\pi_k(t)$ that mark progressively finer approximations to t:

$$X_t - X_{t_0} = (X_{\pi_\kappa(t)} - X_{t_0}) + (X_{\pi_{\kappa+1}(t)} - X_{\pi_\kappa(t)}) + \cdots + (X_t - X_{\pi_K(t)}), \tag{8.8}$$

see Figure 8.2 for illustration. The first and last terms of this sum are zero by

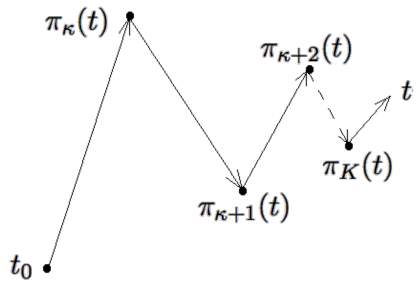


Figure 8.2 Chaining: a walk from a fixed point t_0 to an arbitrary point t in T along elements $\pi_k(T)$ of progressively finer nets of T

(8.6), so we have

$$X_t - X_{t_0} = \sum_{k=\kappa+1}^{K} (X_{\pi_k(t)} - X_{\pi_{k-1}(t)}). \tag{8.9}$$

Since the supremum of the sum is bounded by the sum of suprema, this yields

$$\mathbb{E} \sup_{t \in T} (X_t - X_{t_0}) \le \sum_{k=\kappa+1}^{K} \mathbb{E} \sup_{t \in T} (X_{\pi_k(t)} - X_{\pi_{k-1}(t)}). \tag{8.10}$$

Step 2: Controlling the increments. Although each term in the bound (8.10) still has a supremum over the entire set T, a closer look reveals that it is actually a maximum over a much smaller set, namely the set all possible pairs $(\pi_k(t), \pi_{k-1}(t))$. The number of such pairs is

$$|T_k| \cdot |T_{k-1}| \le |T_k|^2,$$

a number that we can control through (8.5).

Next, for a fixed t, the increments in (8.10) can be bounded as follows:

$$\begin{aligned}\|X_{\pi_k(t)} - X_{\pi_{k-1}(t)}\|_{\psi_2} &\le d(\pi_k(t),\, \pi_{k-1}(t)) \quad \text{(by (8.1) and since } K=1) \\ &\le d(\pi_k(t), t) + d(t,\, \pi_{k-1}(t)) \quad \text{(by triangle inequality)} \\ &\le \varepsilon_k + \varepsilon_{k-1} \quad \text{(by (8.7))} \\ &\le 2\varepsilon_{k-1}.\end{aligned}$$

Recall from Exercise 2.5.10 that the expected maximum of N sub-gaussian random variables is at most $CL\sqrt{\log N}$ where L is the maximal ψ_2 norm. Thus we can bound each term in (8.10) as follows:

$$\mathbb{E} \sup_{t\in T} (X_{\pi_k(t)} - X_{\pi_{k-1}(t)}) \le C\varepsilon_{k-1}\sqrt{\log |T_k|}. \tag{8.11}$$

Step 3: Summing up the increments. We have shown that

$$\mathbb{E} \sup_{t\in T} (X_t - X_{t_0}) \le C \sum_{k=\kappa+1}^{K} \varepsilon_{k-1}\sqrt{\log |T_k|}. \tag{8.12}$$

It remains substitute the values $\varepsilon_k = 2^{-k}$ from (8.4) and the bounds (8.5) on $|T_k|$, and conclude that

$$\mathbb{E} \sup_{t\in T} (X_t - X_{t_0}) \le C_1 \sum_{k=\kappa+1}^{K} 2^{-k}\sqrt{\log \mathcal{N}(T, d, 2^{-k})}.$$

Theorem 8.1.4 is proved. □

Let us now deduce the integral form of Dudley's inequality.

Proof of Dudley's integral inequality, Theorem 8.1.3. To convert the sum (8.2) into an integral, we express 2^{-k} as $2\int_{2^{-k-1}}^{2^{-k}} d\varepsilon$. Then

$$\sum_{k\in\mathbb{Z}} 2^{-k}\sqrt{\log \mathcal{N}(T, d, 2^{-k})} = 2\sum_{k\in\mathbb{Z}} \int_{2^{-k-1}}^{2^{-k}} \sqrt{\log \mathcal{N}(T, d, 2^{-k})}\, d\varepsilon.$$

Within the limits of integral, $2^{-k} \ge \varepsilon$, so $\log \mathcal{N}(T, d, 2^{-k}) \le \log \mathcal{N}(T, d, \varepsilon)$ and the sum is bounded by

$$2\sum_{k\in\mathbb{Z}} \int_{2^{-k-1}}^{2^{-k}} \sqrt{\log \mathcal{N}(T, d, \varepsilon)}\, d\varepsilon = 2\int_0^\infty \sqrt{\log \mathcal{N}(T, d, \varepsilon)}\, d\varepsilon.$$

The proof is complete. □

Remark 8.1.5 (Supremum of increments)**.** A quick glance at the proof reveals that the chaining method actually yields the bound

$$\mathbb{E} \sup_{t\in T} |X_t - X_{t_0}| \le CK \int_0^\infty \sqrt{\log \mathcal{N}(T, d, \varepsilon)}\, d\varepsilon$$

for any fixed $t_0 \in T$. Combining it with a similar bound for $X_s - X_{t_0}$ and using triangle inequality, we deduce that

$$\mathbb{E} \sup_{t,s\in T} |X_t - X_s| \le CK \int_0^\infty \sqrt{\log \mathcal{N}(T,d,\varepsilon)}\, d\varepsilon.$$

Note that in either of these two bounds, we need not require the mean zero assumption $\mathbb{E}\, X_t = 0$. It is required, however, in Dudley's Theorem 8.1.3; otherwise it may fail. (Why?)

Dudley's inequality gives a bound on the expectation only, but adapting the argument yields a nice tail bound as well.

Theorem 8.1.6 (Dudley's integral inequality: tail bound)**.** *Let $(X_t)_{t\in T}$ be a random process on a metric space (T,d) with sub-gaussian increments as in (8.1). Then, for every $u \ge 0$, the event*

$$\sup_{t,s\in T} |X_t - X_s| \le CK \left[\int_0^\infty \sqrt{\log \mathcal{N}(T,d,\varepsilon)}\, d\varepsilon + u \cdot \operatorname{diam}(T) \right]$$

holds with probability at least $1 - 2\exp(-u^2)$.

Exercise 8.1.7. ☕☕☕ Prove Theorem 8.1.6. To this end, first obtain a high-probability version of (8.11):

$$\sup_{t\in T} (X_{\pi_k(t)} - X_{\pi_{k-1}(t)}) \le C\varepsilon_{k-1} \left[\sqrt{\log |T_k|} + z \right]$$

with probability at least $1 - 2\exp(-z^2)$.

Use this inequality with $z = z_k$ to control all such terms simultaneously. Summing them up, deduce a bound on $\sup_{t\in T} |X_t - X_{t_0}|$ with probability at least $1 - 2\sum_k \exp(-z_k^2)$. Finally, choose the values for z_k that give you a good bound; one can set $z_k = u + \sqrt{k-\kappa}$ for example.

Exercise 8.1.8 (Equivalence of Dudley's integral and sum)**.** ☕☕ In the proof of Theorem 8.1.3 we bounded Dudley's sum by the integral. Show the reverse bound:

$$\int_0^\infty \sqrt{\log \mathcal{N}(T,d,\varepsilon)}\, d\varepsilon \le C \sum_{k\in\mathbb{Z}} 2^{-k} \sqrt{\log \mathcal{N}(T,d,2^{-k})}.$$

8.1.1 Remarks and Examples

Remark 8.1.9 (Limits of Dudley's integral)**.** Although Dudley's integral is formally over $[0,\infty]$, we can clearly make the upper bound equal the diameter of T in the metric d, thus

$$\mathbb{E} \sup_{t\in T} X_t \le CK \int_0^{\operatorname{diam}(T)} \sqrt{\log \mathcal{N}(T,d,\varepsilon)}\, d\varepsilon. \tag{8.13}$$

Indeed, if $\varepsilon > \operatorname{diam}(T)$ then a single point (any point in T) is an ε-net of T, which shows that $\log \mathcal{N}(T,d,\varepsilon) = 0$ for such ε.

Let us apply Dudley's inequality for the canonical Gaussian process, just like we did with Sudakov's inequality in Section 7.4.1. We immediately obtain the following bound.

Theorem 8.1.10 (Dudley's inequality for sets in $\mathbb{R}^n$)**.** *For any set $T \subset R^n$, we have*

$$w(T) \leq C \int_0^\infty \sqrt{\log \mathcal{N}(T, \varepsilon)}\, d\varepsilon.$$

Example 8.1.11. Let us test Dudley's inequality for the unit Euclidean ball $T = B_2^n$. Recall from (4.10) that

$$\mathcal{N}(B_2^n, \varepsilon) \leq \left(\frac{3}{\varepsilon}\right)^n \quad \text{for } \varepsilon \in (0, 1]$$

and $\mathcal{N}(B_2^n, \varepsilon) = 1$ for $\varepsilon > 1$. Then Dudley's inequality yields a converging integral

$$w(B_2^n) \leq C \int_0^1 \sqrt{n \log \frac{3}{\varepsilon}}\, d\varepsilon \leq C_1 \sqrt{n}.$$

This is optimal: indeed, as we know from (7.16), the Gaussian width of B_2^n is equivalent to $\sqrt{n}$ up to a constant factor.

Exercise 8.1.12 (Dudley's inequality can be loose)**.** Let $e_1, \ldots, e_n$ denote the canonical basis vectors in $\mathbb{R}^n$. Consider the set

$$T := \left\{ \frac{e_k}{\sqrt{1 + \log k}},\ k = 1, \ldots, n \right\}.$$

(a) Show that

$$w(T) \leq C,$$

where as usual C denotes an absolute constant.

Hint: This should be straightforward from Exercise 2.5.10.

(b) Show that

$$\int_0^\infty \sqrt{\log \mathcal{N}(T, d, \varepsilon)}\, d\varepsilon \to \infty$$

as $n \to \infty$.

Hint: The first m vectors in T form a $(1/\sqrt{\log m})$-separated set.

*8.1.2 * Two-sided Sudakov's inequality*

This sub-section is optional; further material is not based on it.

As we just saw in Exercise 8.1.12, in general there is a gap between Sudakov's and Dudley's inequalities. Fortunately, this gap is only logarithmically large. Let us make this statement more precise and show that Sudakov's inequality in $\mathbb{R}^n$ (Corollary 7.4.3) is optimal up to a $\log n$ factor.

Theorem 8.1.13 (Two-sided Sudakov's inequality)**.** *Let $T \subset \mathbb{R}^n$ and set*

$$s(T) := \sup_{\varepsilon \geq 0} \varepsilon \sqrt{\log \mathcal{N}(T, \varepsilon)}.$$

Then

$$c \cdot s(T) \leq w(T) \leq C \log(n) \cdot s(T).$$

Proof The lower bound is a form of Sudakov's inequality (Corollary 7.4.3). To prove the upper bound, the main idea is that the chaining process converges exponentially fast, and thus $O(\log n)$ steps should suffice to walk from t_0 to somewhere very near t.

As we already noted in (8.13), the coarsest scale in the chaining sum (8.9) can be chosen as the diameter of T. In other words, we can start the chaining at κ which is the smallest integer such that

$$2^{-\kappa} < \operatorname{diam}(T).$$

This is not different from what we did before. What will be different is the finest scale. Instead of going all the way down, let us stop chaining at K which is the largest integer for which

$$2^{-K} \geq \frac{w(T)}{4\sqrt{n}}.$$

(It will be clear why we made this choice in a second.)

Then the last term in (8.8) may not be zero as before, and instead of (8.9) we need to bound

$$w(T) \leq \sum_{k=\kappa+1}^{K} \mathbb{E} \sup_{t \in T} (X_{\pi_k(t)} - X_{\pi_{k-1}(t)}) + \mathbb{E} \sup_{t \in T} (X_t - X_{\pi_K(t)}). \tag{8.14}$$

To control the last term, recall that $X_t = \langle g, t \rangle$ is the canonical process, so

$$\begin{aligned}
\mathbb{E} \sup_{t \in T} (X_t - X_{\pi_K(t)}) &= \mathbb{E} \sup_{t \in T} \langle g, t - \pi_K(t) \rangle \\
&\leq 2^{-K} \cdot \mathbb{E} \|g\|_2 \quad (\text{since } \|t - \pi_K(t)\|_2 \leq 2^{-K}) \\
&\leq 2^{-K} \sqrt{n} \\
&\leq \frac{w(T)}{2\sqrt{n}} \cdot \sqrt{n} \quad (\text{by definition of } K) \\
&\leq \frac{1}{2} w(T).
\end{aligned}$$

Putting this into (8.14) and subtracting $\frac{1}{2} w(T)$ from both sides, we conclude that

$$w(T) \leq 2 \sum_{k=\kappa+1}^{K} \mathbb{E} \sup_{t \in T} (X_{\pi_k(t)} - X_{\pi_{k-1}(t)}). \tag{8.15}$$

Thus, we have removed the last term from (8.14). Each of the remaining terms can be bounded as before. The number of terms in this sum is

$$\begin{aligned} K - \kappa &\le \log_2 \frac{\mathrm{diam}(T)}{w(T)/4\sqrt{n}} \quad \text{(by definition of } K \text{ and } \kappa) \\ &\le \log_2 \left(4\sqrt{n} \cdot \sqrt{2\pi}\right) \quad \text{(by property f of Proposition 7.5.2)} \\ &\le C \log n. \end{aligned}$$

Thus we can replace the sum by the maximum in (8.15) by paying a factor $C \log n$. This completes the argument like before, in the proof of Theorem 8.1.4. □

Exercise 8.1.14 (Limits in Dudley's integral)**.** ☕☕☕ Prove the following improvement of Dudley's inequality (Theorem 8.1.10). For any set $T \subset R^n$, we have

$$w(T) \le C \int_a^b \sqrt{\log \mathcal{N}(T, \varepsilon)}\, d\varepsilon \quad \text{where} \quad a = \frac{c w(T)}{\sqrt{n}}, \quad b = \mathrm{diam}(T).$$

8.2 Application: empirical processes

We give an application of Dudley's inequality to *empirical processes*, which are certain random processes indexed by functions. The theory of empirical processes is a large branch of probability theory, and we only scratch its surface here. Let us consider a motivating example.

8.2.1 Monte-Carlo method

Suppose we want to evaluate the integral of a function $f : \Omega \to \mathbb{R}$ with respect to some probability measure μ on some domain $\Omega \subset \mathbb{R}^d$:

$$\int_\Omega f \, d\mu,$$

see Figure 8.3a. For example, we could be interested in computing $\int_0^1 f(x)\, dx$ for a function $f : [0,1] \to \mathbb{R}$.

We use probability to evaluate this integral. Consider a random point X that takes values in Ω according to the law μ, i.e.

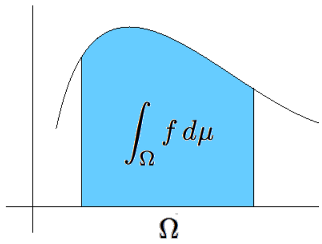
$$\mathbb{P}\left\{X \in A\right\} = \mu(A) \quad \text{for any measurable set } A \subset \Omega.$$

(For example, to evaluate $\int_0^1 f(x)\, dx$, we take $X \sim \mathrm{Unif}[0,1]$.) Then we may interpret the integral as expectation:

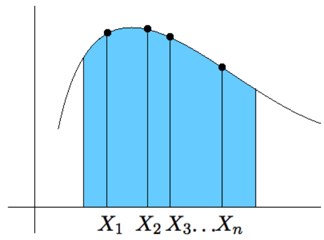
$$\int_\Omega f \, d\mu = \mathbb{E} f(X).$$

Let $X_1, X_2, \ldots$ be i.i.d. copies of X. The law of large numbers (Theorem 1.3.1) yields that

$$\frac{1}{n} \sum_{i=1}^n f(X_i) \to \mathbb{E} f(X) \quad \text{almost surely} \tag{8.16}$$



(a) The problem is to compute the integral of f on a domain Ω.



(b) The integral is approximated by the sum $\frac{1}{n}\sum_1^n f(X_i)$ with randomly sampled points X_i.

Figure 8.3 Monte-Carlo method for randomized, numerical integration.

as $n \to \infty$. This means that we can approximate the integral by the sum

$$\int_\Omega f\,d\mu \approx \frac{1}{n}\sum_{i=1}^n f(X_i) \tag{8.17}$$

where the points X_i are drawn at random from the domain Ω; see Figure 8.3b for illustration. This way of numerically computing integrals is called the *Monte-Carlo method.*

Remark 8.2.1 (Error rate)**.** Note that the average error in (8.17) is $O(1/\sqrt{n})$. Indeed, as we note in (1.5), the rate of convergence in the law of large numbers is

$$\mathbb{E}\left|\frac{1}{n}\sum_{i=1}^n f(X_i) - \mathbb{E} f(X)\right| \le \left[\operatorname{Var}\left(\frac{1}{n}\sum_{i=1}^n f(X_i)\right)\right]^{1/2} = O\Big(\frac{1}{\sqrt{n}}\Big). \tag{8.18}$$

Remark 8.2.2. Note that we do not even need to know the measure μ to evaluate the integral $\int_\Omega f\,d\mu$; it suffices to be able to draw random samples X_i according to μ. Similarly, we do not even need to know f at all points in the domain; a few random points suffice.

8.2.2 A uniform law of large numbers

Can we use the same sample $X_1, \dots, X_n$ to evaluate the integral of *any* function $f : \Omega \to \mathbb{R}$? Of course, not. For a given sample, one can choose a function that oscillates in a the wrong way between the sample points, and the approximation (8.17) will fail.

Will it help if we consider only those functions f that do not oscillate wildly – for example, Lipschitz functions? It will. Our next theorem states that Monte-Carlo method (8.17) does work well simultaneously over the class of Lipschitz functions

$$\mathcal{F} := \{f : [0,1] \to \mathbb{R},\ \|f\|_{\mathrm{Lip}} \le L\}, \tag{8.19}$$

where L is any fixed number.

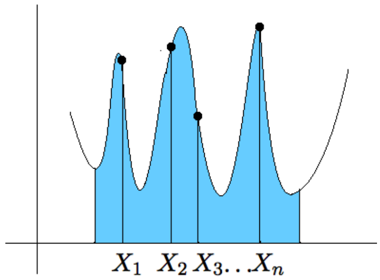


Figure 8.4 One can not use the same sample $X_1, \ldots, X_n$ to approximate the integral of *any* function f.

Theorem 8.2.3 (Uniform law of large numbers)**.** *Let $X, X_1, X_2, \ldots, X_n$ be i.i.d. random variables taking values in* $[0,1]$. *Then*

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^{n} f(X_i) - \mathbb{E} f(X) \right| \leq \frac{CL}{\sqrt{n}}. \tag{8.20}$$

Remark 8.2.4. Before we prove this result, let us pause to emphasize its key point: the supremum over $f \in \mathcal{F}$ appears *inside* the expectation. By Markov's inequality, this means that with high probability, a random sample $X_1, \ldots, X_n$ is good. And "good" means that using this sample, we can approximate the integral of *any* function $f \in \mathcal{F}$ with error bounded by the same quantity $CL/\sqrt{n}$. This is the same rate of convergence the classical Law of Large numbers (8.18) guarantees for a *single* function f. So we paid essentially nothing for making the law of large numbers uniform over the class of functions $\mathcal{F}$.

To prepare for the proof of Theorem 8.2.3, it will be useful to view the left side of (8.20) as the magnitude of a random process indexed by functions $f \in \mathcal{F}$. Such random processes are called *empirical processes*.

Definition 8.2.5. Let $\mathcal{F}$ be a class of real-valued functions $f : \Omega \to \mathbb{R}$ where (Ω, Σ, μ) is a probability space. Let X be a random point in Ω distributed according to the law μ, and let $X_1, X_2, \ldots, X_n$ be independent copies of X. The random process $(X_f)_{f \in \mathcal{F}}$ defined by

$$X_f := \frac{1}{n} \sum_{i=1}^{n} f(X_i) - \mathbb{E} f(X) \tag{8.21}$$

is called an *empirical process* indexed by $\mathcal{F}$.

Proof of Theorem 8.2.3 Without loss of generality, it is enough to prove the theorem for the class

$$\mathcal{F} := \left\{ f : [0,1] \to [0,1],\ \|f\|_{\mathrm{Lip}} \leq 1 \right\}. \tag{8.22}$$

(Why?) We would like to bound the magnitude

$$\mathbb{E} \sup_{f \in \mathcal{F}} |X_f|$$

of the empirical process $(X_f)_{f\in\mathcal{F}}$ defined in (8.21).

Step 1: checking sub-gaussian increments. We can do this using Dudley's inequality, Theorem 8.1.3. To apply this result, we just need to check that the empirical process has sub-gaussian increments. So, fix a pair of functions $f, g \in \mathcal{F}$ and consider

$$\|X_f - X_g\|_{\psi_2} = \frac{1}{n}\Big\|\sum_{i=1}^n Z_i\Big\|_{\psi_2} \quad \text{where} \quad Z_i := (f-g)(X_i) - \mathbb{E}(f-g)(X).$$

Random variables Z_i are independent and have mean zero. So, by Proposition 2.6.1 we have

$$\|X_f - X_g\|_{\psi_2} \lesssim \frac{1}{n}\Big(\sum_{i=1}^n \|Z_i\|_{\psi_2}^2\Big)^{1/2}.$$

Now, using centering (Lemma 2.6.8) we have

$$\|Z_i\|_{\psi_2} \lesssim \|(f-g)(X_i)\|_{\psi_2} \lesssim \|f-g\|_\infty.$$

It follows that

$$\|X_f - X_g\|_{\psi_2} \lesssim \frac{1}{n}\cdot n^{1/2}\|f-g\|_\infty = \frac{1}{\sqrt{n}}\|f-g\|_\infty.$$

Step 2: applying Dudley's inequality. We found that the empirical process $(X_f)_{f\in\mathcal{F}}$ has sub-gaussian increments with respect to the L^∞ norm. This allows us to apply Dudley's inequality. Note that (8.22) implies that the diameter of $\mathcal{F}$ in L^∞ metric is bounded by 1. Thus

$$\mathbb{E}\sup_{f\in\mathcal{F}}|X_f| = \mathbb{E}\sup_{f\in\mathcal{F}}|X_f - X_0| \lesssim \frac{1}{\sqrt{n}}\int_0^1 \sqrt{\log\mathcal{N}(\mathcal{F}, \|\cdot\|_\infty, \varepsilon)}\,d\varepsilon.$$

(Here we used that the zero function belongs to $\mathcal{F}$ and used the version of Dudley's inequality from Remark 8.1.5; see also (8.13)).

Using that all functions in $f \in \mathcal{F}$ are Lipschitz with $\|f\|_{\mathrm{Lip}} \le 1$, it is not difficult to bound the covering numbers of $\mathcal{F}$ as follows:

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_\infty, \varepsilon) \le \Big(\frac{C}{\varepsilon}\Big)^{C/\varepsilon};$$

we will show this in Exercise 8.2.6 below. This bound makes Dudley's integral converge, and we conclude that

$$\mathbb{E}\sup_{f\in\mathcal{F}}|X_f| \lesssim \frac{1}{\sqrt{n}}\int_0^1 \sqrt{\frac{C}{\varepsilon}\log\frac{C}{\varepsilon}}\,d\varepsilon \lesssim \frac{1}{\sqrt{n}}.$$

Theorem 8.2.3 is proved. □

Exercise 8.2.6 (Metric entropy of the class of Lipschitz functions). ☕☕☕ Consider the class of functions

$$\mathcal{F} := \{f : [0,1] \to [0,1],\ \|f\|_{\mathrm{Lip}} \le 1\}.$$

Show that

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_\infty, \varepsilon) \le \left(\frac{2}{\varepsilon}\right)^{2/\varepsilon} \quad \text{for any } \varepsilon \in (0,1).$$

Hint: Put a mesh on the square $[0,1]^2$ with step ε. Given $f \in \mathcal{F}$, show that $\|f - f_0\|_\infty \le \varepsilon$ for some function f_0 whose graph follows the mesh; see Figure 8.5. The number all mesh-following functions f_0 is bounded by $(1/\varepsilon)^{1/\varepsilon}$. Next, use the result of Exercise 4.2.9.

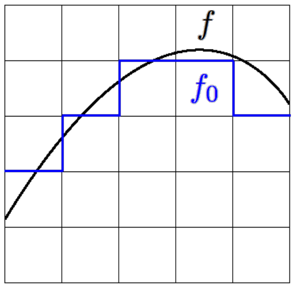


Figure 8.5 Bounding the metric entropy of the class of Lipschitz functions in Exercise 8.2.6. A Lipschitz function f is approximated by a function f_0 on a mesh.

Exercise 8.2.7 (An improved bound on the metric entropy)**.** ☕☕☕ Improve the bound in Exercise 8.2.6 to

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_\infty, \varepsilon) \le e^{C/\varepsilon} \quad \text{for any } \varepsilon > 0.$$

Hint: Use that f is Lipschitz to find a better bound on the number of possible functions f_0.

Exercise 8.2.8 (Higher dimensions)**.** ☕☕☕ Let $[0,1]^d$ be the unit cube in dimension $d \ge 1$ equipped with the $\|\cdot\|_\infty$ metric. Consider the class of functions

$$\mathcal{F} := \left\{ f : [0,1]^d \to \mathbb{R},\ f(0) = 0, \|f\|_{\mathrm{Lip}} \le 1 \right\}.$$

Show that

$$\mathcal{N}(\mathcal{F}, \|\cdot\|_\infty, \varepsilon) \le e^{C/\varepsilon^d} \quad \text{for any } \varepsilon > 0.$$

8.2.3 Empirical measure

Let us take one more look at the Definition 8.2.5 of empirical processes. Consider a probability measure μ_n that is uniformly distributed on the sample $X_1, \dots, X_n$, that is

$$\mu_n(\{X_i\}) = \frac{1}{n} \quad \text{for every } i = 1, \dots, n. \tag{8.23}$$

Note that μ_n is a *random* measure. It is called the *empirical measure.*

While the integral of f with respect to the original measure μ is the $\mathbb{E} f(X)$ (the "population" average of f) the integral of f with respect to the empirical measure is $\frac{1}{n}\sum_{i=1}^n f(X_i)$ (the "sample", or empirical, average of f). In the literature on empirical processes, the population expectation of f is denoted by μf, and the empirical expectation, by $\mu_n f$:

$$\mu f = \int f \, d\mu = \mathbb{E} f(X), \qquad \mu_n f = \int f \, d\mu_n = \frac{1}{n}\sum_{i=1}^n f(X_i).$$

The empirical process X_f in (8.21) thus measures the deviation of population expectation from the empirical expectation:

$$X_f = \mu f - \mu_n f.$$

The uniform law of large numbers (8.20) gives a uniform bound on the deviation

$$\mathbb{E} \sup_{f \in \mathcal{F}} |\mu_n f - \mu f| \tag{8.24}$$

over the class of Lipschitz functions $\mathcal{F}$ defined in (8.19).

The quantity (8.24) can be thought as a distance between the measures μ_n and μ. It is called the *Wasserstein's distance* $W_1(\mu, \mu_n)$. The Wasserstein distance has an equivalent formulation as the *transportation cost* of measure μ into measure μ_n, where the cost of moving a mass (probability) $p > 0$ is proportional to p and to the distance moved. The equivalence between the transportation cost and (8.24) is provided by Kantorovich-Rubinstein's duality theorem.

8.3 VC dimension

In this section, we introduce the notion of VC dimension, which plays a major role in statistical learning theory. We relate VC dimension to covering numbers, and then, through Dudley's inequality, to random processes and uniform law of large numbers. Applications to statistical learning theory will be given in next section.

8.3.1 Definition and examples

VC-dimension is a measure of complexity of classes of Boolean functions. By a class of Boolean functions we mean any collection $\mathcal{F}$ of functions $f : \Omega \to \{0, 1\}$ defined on a common domain Ω.

Definition 8.3.1 (VC dimension)**.** Consider a class $\mathcal{F}$ of Boolean functions on some domain Ω. We say that a subset $\Lambda \subseteq \Omega$ is *shattered* by $\mathcal{F}$ if any function $g : \Lambda \to \{0, 1\}$ can be obtained by restricting some function $f \in \mathcal{F}$ onto Λ. The *VC dimension* of $\mathcal{F}$, denoted $\mathrm{vc}(\mathcal{F})$, is the largest cardinality[1] of a subset $\Lambda \subseteq \Omega$ shattered by $\mathcal{F}$.

The definition of VC dimension may take some time to fully comprehend. We work out a few examples to illustrate this notion.

[1] If the largest cardinality does not exist, we set $\mathrm{vc}(\mathcal{F}) = \infty$.

Example 8.3.2 (Intervals)**.** Let $\mathcal{F}$ be the class of indicators of all closed intervals in $\mathbb{R}$, that is

$$\mathcal{F} := \left\{ \mathbf{1}_{[a,b]} : \ a, b \in \mathbb{R}, \ a \leq b \right\}.$$

We claim that there exists a two-point set $\Lambda \subset \mathbb{R}$ that is shattered by $\mathcal{F}$, and thus

$$\mathrm{vc}(\mathcal{F}) \geq 2.$$

Take, for example, $\Lambda := \{3, 5\}$. It is easy to see that each of the four possible functions $g : \Lambda \to \{0, 1\}$ is a restriction of some indicator function $f = \mathbf{1}_{[a,b]}$ onto Λ. For example, the function g defined by $g(3) = 1$, $g(5) = 0$ is a restriction of $f = \mathbf{1}_{[2,4]}$ onto Λ, since $f(3) = g(3) = 1$ and $f(5) = g(5) = 0$. The three other possible functions g can be treated similarly; see Figure 8.6. Thus $\Lambda = \{3, 5\}$ is indeed shattered by $\mathcal{F}$, as claimed.

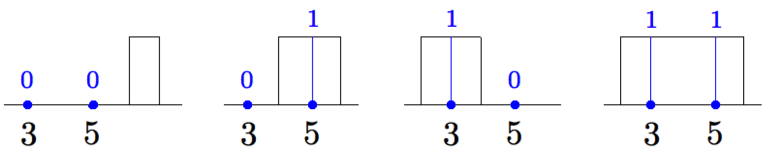


Figure 8.6 The function $g(3) = g(5) = 0$ is a restriction of $\mathbf{1}_{[6,7]}$ onto $\Lambda = \{3, 5\}$ (left). The function $g(3) = 0$, $g(5) = 1$ is a restriction of $\mathbf{1}_{[4,6]}$ onto Λ (middle left). The function $g(3) = 1$, $g(5) = 0$ is a restriction of $\mathbf{1}_{[2,4]}$ onto Λ (middle right). The function $g(3) = g(5) = 1$ is a restriction of $\mathbf{1}_{[2,6]}$ onto Λ (right).

Next, we claim that no three-point set $\Lambda = \{p, q, r\}$ can be shattered by $\mathcal{F}$, and thus

$$\mathrm{vc}(\mathcal{F}) = 2.$$

To see this, assume $p < q < r$ and define the function $g : \Lambda \to \{0, 1\}$ by $g(p) = 1$, $g(q) = 0$, $g(r) = 1$. Then g can not be a restriction of any indicator $\mathbf{1}_{[a,b]}$ onto Λ, for otherwise $[a, b]$ must contain two points p and r but not the point q that lies between them, which is impossible.

Example 8.3.3 (Half-planes)**.** Let $\mathcal{F}$ be the class of indicators of all closed half-planes in $\mathbb{R}^2$. We claim that there is a three-point set $\Lambda \subset \mathbb{R}^2$ that is shattered by $\mathcal{F}$, and thus

$$\mathrm{vc}(\mathcal{F}) \geq 3.$$

To see this, let Λ be a set of three points in general position, such as in Figure 8.7. Then each of the $2^3 = 8$ functions $g : \Lambda \to \{0, 1\}$ is a restriction of the indicator function of some half-plane. To see this, arrange the half-plane to contain exactly those points of Λ where g takes value 1, which can always be done – see Figure 8.7.

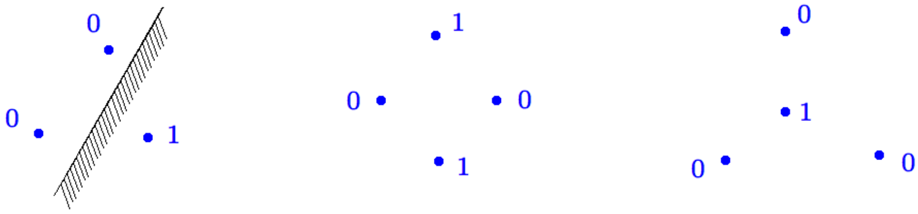


Figure 8.7 Left: a three-point set Λ and function $g : \Lambda \to \{0,1\}$ (values shown in blue). Such g is a restriction of the indicator function of the shaded half-plane. Middle and right: two kinds of four-points sets Λ in general position, and functions $g : \Lambda \to \{0,1\}$. In each case, no half-plane can contain exactly the points with value 1. Thus, g is not a restriction of the indicator function of any half-plane.

Next, we claim that no four-point set can be shattered by $\mathcal{F}$, and thus

$$\mathrm{vc}(\mathcal{F}) = 3.$$

There are two possible arrangements of the four-point sets Λ in general position, shown in Figure 8.7. (What if Λ is not in general position? Analyze this case.) In each of the two cases, there exists a 0/1 labeling of the points 0 and 1 such that no half-plane can contain exactly the points labeled 1; see Figure 8.7. This means that in each case, there exists a function $g : \Lambda \to \{0,1\}$ that is not a restriction of any function $f \in \mathcal{F}$ onto Λ, and thus Λ is not shattered by $\mathcal{F}$ as claimed.

Example 8.3.4. Let $\Omega = \{1,2,3\}$. We can conveniently represent Boolean functions on Ω as binary strings of length three. Consider the class

$$\mathcal{F} := \{001, 010, 100, 111\}.$$

The set $\Lambda = \{1,3\}$ is shattered by $\mathcal{F}$. Indeed, restricting the functions in $\mathcal{F}$ onto Λ amounts to dropping the second digit, thus producing strings $00, 01, 10, 11$. Thus, the restriction produces all possible binary strings of length two, or equivalently, all possible functions $g : \Lambda \to \{0,1\}$. Hence Λ is shattered by $\mathcal{F}$, and thus $\mathrm{vc}(\mathcal{F}) \geq |\Lambda| = 2$. On the other hand, the (only) three-point set $\{1,2,3\}$ is not shattered by $\mathcal{F}$, as this would require all eight binary digits of length three to appear in $\mathcal{F}$, which is not true.

Exercise 8.3.5 (Pairs of intervals)**.** ☕☕ Let $\mathcal{F}$ be the class of indicators of sets of the form $[a,b] \cup [c,d]$ in $\mathbb{R}$. Show that

$$\mathrm{vc}(\mathcal{F}) = 4.$$

Exercise 8.3.6 (Circles)**.** ☕☕☕ Let $\mathcal{F}$ be the class of indicators of all circles in $\mathbb{R}^2$. Show that

$$\mathrm{vc}(\mathcal{F}) = 3.$$

Exercise 8.3.7 (Rectangles)**.** ☕☕☕ Let $\mathcal{F}$ be the class of indicators of all closed axis-aligned rectangles, i.e. product sets $[a, b] \times [c, d]$, in $\mathbb{R}^2$. Show that

$$\mathrm{vc}(\mathcal{F}) = 4.$$

Exercise 8.3.8 (Squares)**.** ☕☕☕ Let $\mathcal{F}$ be the class of indicators of all closed axis-aligned squares, i.e. product sets $[a, a+d] \times [b, b+d]$, in $\mathbb{R}^2$. Show that

$$\mathrm{vc}(\mathcal{F}) = 3.$$

Exercise 8.3.9 (Polygons)**.** ☕☕☕ Let $\mathcal{F}$ be the class of indicators of all convex polygons in $\mathbb{R}^2$, without any restriction on the number of vertices. Show that

$$\mathrm{vc}(\mathcal{F}) = \infty.$$

Remark 8.3.10 (VC dimension of classes of sets)**.** We may talk about VC dimension of classes of *sets* instead of functions. This is due to the natural correspondence between the two: a Boolean function f on Ω determines the subset $\{x \in \Omega : \ f(x) = 1\}$, and, vice versa, a subset $\Omega_0 \subset \Omega$ determines the Boolean function $f = \mathbf{1}_{\Omega_0}$. In this language, the VC dimension of the set of intervals in $\mathbb{R}$ equals 2, the VC dimension of the set of half-planes in $\mathbb{R}^2$ equals 3, and so on.

Exercise 8.3.11. ☕ Give the definition of VC dimension of a class of subsets of Ω without mentioning any functions.

Remark 8.3.12 (More examples)**.** It can be shown that the VC dimension of the class of all rectangles on the plane (not necessarily axis-aligned) equals 7. For the class of all polygons with k vertices on the plane, the VC dimension is also $2k + 1$. For the class of half-spaces in $\mathbb{R}^n$, the VC dimension is $n + 1$.

8.3.2 Pajor's Lemma

Consider a class of Boolean functions $\mathcal{F}$ on a *finite* set Ω. We study a remarkable connection between the cardinality $|\mathcal{F}|$ and VC dimension of $\mathcal{F}$. Somewhat oversimplifying, we can say that $|\mathcal{F}|$ is exponential in $\mathrm{vc}(\mathcal{F})$. A lower bound is trivial:

$$|\mathcal{F}| \geq 2^{\mathrm{vc}(\mathcal{F})}.$$

(Check!) We now pass to upper bounds; they are less trivial. The following lemma states that there are as many shattered subsets of Ω as the functions in $\mathcal{F}$.

Lemma 8.3.13 (Pajor's Lemma)**.** *Let $\mathcal{F}$ be a class of Boolean functions on a finite set Ω. Then*

$$|\mathcal{F}| \leq \left| \left\{ \Lambda \subseteq \Omega : \ \Lambda \textit{ is shattered by } \mathcal{F} \right\} \right|.$$

We include the empty set $\Lambda = \emptyset$ in the counting on the right side.

Before we prove Pajor's lemma, let us pause to give a quick illustration using Example 8.3.4. There $|\mathcal{F}| = 4$ and there are six subsets Λ that are shattered by $\mathcal{F}$, namely $\{1\}$, $\{2\}$, $\{3\}$, $\{1, 2\}$, $\{1, 3\}$ and $\{2, 3\}$. (Check!) Thus the inequality in Pajor's lemma reads $4 \leq 6$ in this case.

Proof of Pajor's Lemma 8.3.13. We proceed by induction on the cardinality of Ω. The case $|\Omega| = 1$ is trivial, since we include the empty set in the counting. Assume the lemma holds for any n-point set Ω, and let us prove it for Ω with $|\Omega| = n + 1$.

Chopping out one (arbitrary) point from the set Ω, we can express it as

$$\Omega = \Omega_0 \cup \{x_0\}, \quad \text{where} \quad |\Omega_0| = n.$$

The class $\mathcal{F}$ then naturally breaks into two sub-classes

$$\mathcal{F}_0 := \{f \in \mathcal{F} : \ f(x_0) = 0\} \quad \text{and} \quad \mathcal{F}_1 := \{f \in \mathcal{F} : \ f(x_0) = 1\}.$$

By the induction hypothesis, the counting function

$$S(\mathcal{F}) = \left|\left\{\Lambda \subseteq \Omega : \ \Lambda \text{ is shattered by } \mathcal{F}\right\}\right|$$

satisfies[2]

$$S(\mathcal{F}_0) \geq |\mathcal{F}_0| \quad \text{and} \quad S(\mathcal{F}_1) \geq |\mathcal{F}_1|. \tag{8.25}$$

To complete the proof, all we need to check is

$$S(\mathcal{F}) \geq S(\mathcal{F}_0) + S(\mathcal{F}_1), \tag{8.26}$$

for then (8.25) would give $S(\mathcal{F}) \geq |\mathcal{F}_0| + |\mathcal{F}_1| = |\mathcal{F}|$, as needed.

Inequality (8.26) may seem trivial. Any set Λ that is shattered by $\mathcal{F}_0$ or $\mathcal{F}_1$ is automatically shattered by the larger class $\mathcal{F}$, and thus each set Λ counted by $S(\mathcal{F}_0)$ or $S(\mathcal{F}_1)$ is automatically counted by $S(\mathcal{F})$. The problem, however, lies in the double counting. Assume the same set Λ is shattered by *both* $\mathcal{F}_0$ and $\mathcal{F}_1$. The counting function $S(\mathcal{F})$ will not count Λ twice. However, a different set will be counted by $S(\mathcal{F})$, which was not counted by either $S(\mathcal{F}_0)$ or $S(\mathcal{F}_1)$ – namely, $\Lambda \cup \{x_0\}$. A moment's thought reveals that this set is indeed shattered by $\mathcal{F}$. (Check!) This establishes inequality (8.26) and completes the proof of Pajor's Lemma. □

It may be helpful to illustrate the key point in the proof of Pajor's lemma with a specific example.

Example 8.3.14. Let us again go back to Example 8.3.4. Following the proof of Pajor's lemma, we chop out $x_0 = 3$ from $\Omega = \{1, 2, 3\}$, making $\Omega_0 = \{1, 2\}$. The class $\mathcal{F} = \{001, 010, 100, 111\}$ then breaks into two sub-classes

$$\mathcal{F}_0 = \{010, 100\} \quad \text{and} \quad \mathcal{F}_1 = \{001, 111\}.$$

There are exactly two subsets Λ shattered by $\mathcal{F}_0$, namely $\{1\}$ and $\{2\}$, and *the same* subsets are shattered by $\mathcal{F}_1$, making $S(\mathcal{F}_0) = S(\mathcal{F}_1) = 2$. Of course, the same two subsets are also shattered by $\mathcal{F}$, but we need two more shattered subsets to make $S(\mathcal{F}) \geq 4$ for the key inequality (8.26). Here is how we construct them: append $x_0 = 3$ to the already counted subsets Λ. The resulting sets $\{1, 3\}$ and

[2] To properly use the induction hypothesis here, restrict the functions in $\mathcal{F}_0$ and $\mathcal{F}_1$ onto the n-point set Ω_0.

$\{2,3\}$ are also shattered by $\mathcal{F}$, and we have not counted them yet. Now have at least *four* subsets shattered by $\mathcal{F}$, making the key inequality (8.26) in the proof Pajor's lemma true.

Exercise 8.3.15 (Sharpness of Pajor's Lemma)**.** ☕☕ Show that Pajor's Lemma 8.3.13 is sharp.

Hint: Consider the set $\mathcal{F}$ of binary strings of length n with at most d ones. This set is called *Hamming cube.*

8.3.3 Sauer-Shelah Lemma

We now deduce a remarkable upper bound on the cardinality of a function class in terms of VC dimension.

Theorem 8.3.16 (Sauer-Shelah Lemma)**.** *Let $\mathcal{F}$ be a class of Boolean functions on an n-point set Ω. Then*

$$|\mathcal{F}| \le \sum_{k=0}^{d} \binom{n}{k} \le \left(\frac{en}{d}\right)^d$$

where $d = \mathrm{vc}(\mathcal{F})$.

Proof Pajor's Lemma states that $|\mathcal{F}|$ is bounded by the number of subsets $\Lambda \subseteq \Omega$ that are shattered by $\mathcal{F}$. The cardinality of each such set Λ is bounded by $d = \mathrm{vc}(\mathcal{F})$, according to the definition of VC dimension. Thus

$$|\mathcal{F}| \le \left|\left\{\Lambda \subseteq \Omega : \ |\Lambda| \le d\right\}\right| = \sum_{k=0}^{d} \binom{n}{k}$$

since the sum in right hand side gives the total number of subsets of an n-element set with cardinalities at most k. This proves the first inequality of Sauer-Shelah Lemma. The second inequality follows from the bound on the binomial sum we proved in Exercise 0.0.5. □

Exercise 8.3.17 (Sharpness of Sauer-Shelah Lemma)**.** ☕☕ Show that Sauer-Shelah lemma is sharp for all n and d.

Hint: Consider Hamming cube from Exercise 8.3.15.

8.3.4 Covering numbers via VC dimension

Sauer-Shelah Lemma is sharp, but it can only be used for finite function classes $\mathcal{F}$. What about *infinite* function classes $\mathcal{F}$, for example the indicator functions of half-planes in Example 8.3.3? It turns out that we can always bound the *covering numbers* of $\mathcal{F}$ in terms of VC dimension.

Let $\mathcal{F}$ be a class of Boolean functions on a set Ω as before, and let μ be any probability measure on Ω. Then $\mathcal{F}$ can be considered as a metric space under the

$L^2(\mu)$ norm, with the metric on $\mathcal{F}$ given by

$$d(f,g) = \|f-g\|_{L^2(\mu)} = \left(\int_\Omega |f-g|^2 \, d\mu \right)^{1/2}, \quad f,g \in \mathcal{F}.$$

Then we can talk about covering numbers of the class $\mathcal{F}$ in the $L^2(\mu)$ norm, which we denote[3] $\mathcal{N}(\mathcal{F}, L^2(\mu), \varepsilon)$.

Theorem 8.3.18 (Covering numbers via VC dimension)**.** *Let $\mathcal{F}$ be a class of Boolean functions on a probability space (Ω, Σ, μ). Then, for every $\varepsilon \in (0,1)$, we have*

$$\mathcal{N}(\mathcal{F}, L^2(\mu), \varepsilon) \le \left(\frac{2}{\varepsilon}\right)^{Cd}$$

where $d = \mathrm{vc}(\mathcal{F})$.

This result should be compared to the volumetric bound (4.10), which also states that the covering numbers scale exponentially with the dimension. The important difference is that the VC dimension captures a combinatorial rather than linear algebraic complexity of sets.

For a first attempt at proving Theorem 8.3.18, let us assume for a moment that Ω is finite, say $|\Omega| = n$. Then Sauer-Shelah Lemma (Theorem 8.3.16) yields

$$\mathcal{N}(\mathcal{F}, L^2(\mu), \varepsilon) \le |\mathcal{F}| \le \left(\frac{en}{d}\right)^d.$$

This is not quite what Theorem 8.3.18 claims, but it comes close. To improve the bound, we need to remove the dependence on the size n of Ω. Can we reduce the domain Ω to a much smaller subset without harming the covering numbers? It turns out that we can; this will be based on the following lemma.

Lemma 8.3.19 (Dimension reduction)**.** *Let $\mathcal{F}$ be a class of N Boolean functions on a probability space (Ω, Σ, μ). Assume that all functions in $\mathcal{F}$ are ε-separated, that is*

$$\|f-g\|_{L^2(\mu)} > \varepsilon \quad \text{for all distinct } f,g \in \mathcal{F}.$$

Then there exist a number $n \le C\varepsilon^{-4} \log N$ and an n-point set $\Omega_n \subset \Omega$ such that the restrictions of the functions $f \in \mathcal{F}$ onto Ω_n are all distinct.

Proof Our argument will be based on the probabilistic method. We choose the subset Ω_n at random and show that it satisfies the conclusion of the theorem with positive probability. This will automatically imply the existence of at least one suitable choice of Ω_n.

[3] If you are not completely comfortable with measure theory, it may be helpful to consider a discrete case, which is all we need for applications in the next section. Let Ω be an N-point set, say $\Omega = \{1, \dots, N\}$ and μ be the uniform measure on Ω, thus $\mu(i) = 1/N$ for every $i = 1, \dots, N$. In this case, the $L^2(\mu)$ norm of a function $f : \Omega \to \mathbb{R}$ is simply $\|f\|_{L^2(\mu)} = (\frac{1}{N}\sum_{i=1}^N f(i)^2)^{1/2}$. Equivalently, one can think of f as a vector in $\mathbb{R}^N$. The $L^2(\mu)$ is just the scaled Euclidean norm $\|\cdot\|_2$ on $\mathbb{R}^n$, i.e. $\|f\|_{L^2(\mu)} = (1/\sqrt{N})\|f\|_2$.

Let $X, X_1, \ldots, X_n$ independent be random points in Ω distributed[4] according to the law μ. Define the *empirical probability measure* μ_n on Ω by assigning each point X_i probability $1/n$, allowing multiplicities. We would like to show that with positive probability the following event holds:

$$\|f-g\|_{L^2(\mu_n)}^2 := \frac{1}{n}\sum_{i=1}^{n}(f-g)(X_i)^2 > 0 \quad \text{for all distinct} f, g \in \mathcal{F}.$$

This would guarantee that the restrictions of the functions $f \in \mathcal{F}$ onto $\Omega_n := \{X_1, \ldots, X_n\}$ are all distinct.[5]

Fix a pair of distinct functions $f, g \in \mathcal{F}$ and denote $h := (f-g)^2$ for convenience. We would like to bound the deviation

$$\|f-g\|_{L^2(\mu_n)}^2 - \|f-g\|_{L^2(\mu)}^2 = \frac{1}{n}\sum_{i=1}^{n} h(X_i) - \mathbb{E}\, h(X).$$

We have a sum of independent random variables on the right, and we use general Hoeffding's inequality to bound it. To do this, we first check that these random variables are subgaussian. Indeed,[6]

$$\begin{aligned}
\|h(X_i) - \mathbb{E}\, h(X)\|_{\psi_2} &\lesssim \|h(X)\|_{\psi_2} \quad \text{(by Centering Lemma 2.6.8)} \\
&\lesssim \|h(X)\|_\infty \quad \text{(by (2.17))} \\
&\le 1 \quad \text{(since } h = f-g \text{ with } f, g \text{ Boolean).}
\end{aligned}$$

Then general Hoeffding's inequality (Theorem 2.6.2) gives

$$\mathbb{P}\left\{ \left| \|f-g\|_{L^2(\mu_n)}^2 - \|f-g\|_{L^2(\mu)}^2 \right| > \frac{\varepsilon^2}{4} \right\} \le 2\exp(-cn\varepsilon^4).$$

(Check!) Therefore, with probability at least $1 - 2\exp(-cn\varepsilon^4)$, we have

$$\|f-g\|_{L^2(\mu_n)}^2 \ge \|f-g\|_{L^2(\mu)}^2 - \frac{\varepsilon^2}{4} \ge \varepsilon^2 - \frac{\varepsilon^2}{4} = \frac{3\varepsilon^2}{4}, \tag{8.27}$$

where we used triangle inequality and the assumption of the lemma.

This is a good bound, and even stronger than we need, but we proved it for a *fixed* pair $f, g \in \mathcal{F}$ so far. Let us take a union bound over all such pairs; there are at most N^2 of them. Then, with probability at least

$$1 - N^2 \cdot 2\exp(-cn\varepsilon^4), \tag{8.28}$$

the lower bound (8.27) holds simultaneously for all pairs of distinct functions $f, g \in \mathcal{F}$. We can make (8.28) positive by choosing $n := \lceil C'\varepsilon^{-4}\log N \rceil$ with a sufficiently large absolute constant C. Thus the random set Ω_n satisfies the conclusion of the lemma with positive probability. □

[4] For example, if $\Omega = \{1, \ldots, N\}$ then X is a random variable which takes values $1, \ldots, N$ with probability $1/N$ each.

[5] Due to possible multiplicities among the points X_i, the random set Ω_n may contain fewer than n points. If this happens, we can add arbitrarily chosen points to Ω_n to bring its cardinality to n.

[6] The inequalities "$\lesssim$" below hide absolute constant factors.

Proof of Theorem 8.3.18 Let us choose

$$N \geq \mathcal{N}(\mathcal{F}, L^2(\mu), \varepsilon)$$

ε-separated functions in $\mathcal{F}$. (To see why they exist, recall the covering-packing relationship in Lemma 4.2.8.) Apply Lemma 8.3.19 to those functions. We obtain a subset $\Omega_n \subset \Omega$ with

$$|\Omega_n| = n \leq C\varepsilon^{-4} \log N$$

such that the restrictions of those functions onto Ω_n are still $\varepsilon/2$-separated in $L^2(\mu_n)$. We use a much weaker fact – that these restrictions are just distinct. Summarizing, we have a class $\mathcal{F}_n$ of distinct Boolean functions on Ω_n, obtained as restrictions of certain functions from $\mathcal{F}$.

Apply Sauer-Shelah Lemma (Theorem 8.3.16) for $\mathcal{F}_n$. It gives

$$N \leq \left(\frac{en}{d_n}\right)^{d_n} \leq \left(\frac{C\varepsilon^{-4}\log N}{d_n}\right)^{d_n}$$

where $d_n = \mathrm{vc}(\mathcal{F}_n)$. Simplifying this bound,[7] we conclude that

$$N \leq (C\varepsilon^{-4})^{2d_n}.$$

To complete the proof, replace $d_n = \mathrm{vc}(\mathcal{F}_n)$ in this bound by the larger quantity $d = \mathrm{vc}(\mathcal{F})$. □

Remark 8.3.20 (Johnson-Lindenstrauss Lemma for coordinate projections)**.** You may spot a similarity between Dimension Reduction Lemma 8.3.19 and another dimension reduction result, Johnson-Lindenstrauss Lemma (Theorem 5.3.1). Both results state that a random projection of a set of N points onto a dimension of subspace $\log N$ preserves the geometry of the set. The difference is in the distribution of the random subspace. In Johnson-Lindenstrauss Lemma, it is uniformly distributed in the Grassmanian, and in Lemma 8.3.19 it is a coordinate subspace.

Exercise 8.3.21 (Dimension reduction for covering numbers)**.** ☕☕ Let $\mathcal{F}$ be a class of functions on a probability space (Ω, Σ, μ), which are all bounded by 1 in absolute value. Let $\varepsilon \in (0, 1)$. Show that there exists a number $n \leq C\varepsilon^{-4} \log \mathcal{N}(\mathcal{F}, L^2(\mu), \varepsilon)$ and an n-point subset $\Omega_n \subset \Omega$ such that

$$\mathcal{N}(\mathcal{F}, L^2(\mu), \varepsilon) \leq \mathcal{N}(\mathcal{F}, L^2(\mu_n), \varepsilon/4)$$

where μ_n denotes the uniform probability measure on Ω_n.

Hint: Argue as in Lemma 8.3.19 and then use the covering-packing relationship from Lemma 4.2.8.

Exercise 8.3.22. ☕☕ Theorem 8.3.18 is stated for $\varepsilon \in (0, 1)$. What bound holds for larger ε?

[7] To do this, note that $\frac{\log N}{2d_n} = \log(N^{1/2d_n}) \leq N^{1/2d_n}$.

8.3.5 Empirical processes via VC dimension

Let us turn again to the concept of empirical processes that we first introduced in Section 8.2.2. There we showed how to control one specific example of an empirical process, namely the process on the class of Lipschitz functions. In this section we develop a general bound for an arbitrary class Boolean functions.

Theorem 8.3.23 (Empirical processes via VC dimension)**.** *Let $\mathcal{F}$ be a class of Boolean functions on a probability space (Ω, Σ, μ) with finite VC dimension* $\mathrm{vc}(\mathcal{F}) \ge 1$. *Let $X, X_1, X_2, \ldots, X_n$ be independent random points in Ω distributed according to the law μ. Then*

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^{n} f(X_i) - \mathbb{E} f(X) \right| \le C \sqrt{\frac{\mathrm{vc}(\mathcal{F})}{n}}. \tag{8.29}$$

We can quickly derive this result from Dudley's inequality combined with the bound on the covering numbers we just proved in Section 8.3.4. To carry out this argument, it would be helpful to preprocess the empirical process using symmetrization.

Exercise 8.3.24 (Symmetrization for empirical processes)**.** ☕☕ Let $\mathcal{F}$ be a class of functions on a probability space (Ω, Σ, μ). Let $X, X_1, X_2, \ldots, X_n$ be random points in Ω distributed according to the law μ. Prove that

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^{n} f(X_i) - \mathbb{E} f(X) \right| \le 2\, \mathbb{E} \sup_{f \in \mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^{n} \varepsilon_i f(X_i) \right|$$

where $\varepsilon_1, \varepsilon_2, \ldots$ are independent symmetric Bernoulli random variables (which are also independent of $X_1, X_2, \ldots$).

Hint: Modify the proof of Symmetrization Lemma 6.4.2.

Proof of Theorem 8.3.23 First we use symmetrization and bound the left hand side of (8.29) by

$$\frac{2}{\sqrt{n}}\, \mathbb{E} \sup_{f \in \mathcal{F}} |Z_f| \quad \text{where} \quad Z_f := \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \varepsilon_i f(X_i).$$

Next we condition on (X_i), leaving all randomness in the random signs (ε_i). We are going to use Dudley's inequality to bound the process $(Z_f)_{f \in \mathcal{F}}$. For simplicity, let us drop the absolute values for Z_f for a moment; we will deal with this minor issue in Exercise 8.3.25.

To apply Dudley's inequality, we need to check that the increments of the process $(Z_f)_{f \in \mathcal{F}}$ are sub-gaussian. These are

$$\|Z_f - Z_g\|_{\psi_2} = \frac{1}{\sqrt{n}} \left\| \sum_{i=1}^{n} \varepsilon_i (f-g)(X_i) \right\|_{\psi_2} \lesssim \left[\frac{1}{n} \sum_{i=1}^{n} (f-g)(X_i)^2 \right]^{1/2}.$$

Here we used Proposition 2.6.1 and the obvious fact that $\|\varepsilon_i\|_{\psi_2} \lesssim 1$.[8] We can

[8] Keep in mind that here X_i and thus $(f-g)(X_i)$ are fixed numbers due to conditioning.

interpret the last expression as the $L^2(\mu_n)$ norm of the function $f - g$, where μ_n is the uniform probability measure supported on the subset $\{X_1, \dots, X_n\} \subset \Omega$.[9] In other words, the increments satisfy

$$\|Z_f - Z_g\|_{\psi_2} \lesssim \|f - g\|_{L^2(\mu_n)}.$$

Now we can use Dudley's inequality (Theorem 8.1.3) conditionally on (X_i) and get[10]

$$\frac{2}{\sqrt{n}}\,\mathbb{E}\sup_{f\in\mathcal{F}} Z_f \lesssim \frac{1}{\sqrt{n}}\,\mathbb{E}\int_0^1 \sqrt{\log \mathcal{N}(\mathcal{F}, L^2(\mu_n), \varepsilon)}\, d\varepsilon. \tag{8.30}$$

The expectation in the right hand side is obviously with respect to (X_i).

Finally, we use Theorem 8.3.18 to bound the covering numbers:

$$\log \mathcal{N}(\mathcal{F}, L^2(\mu_n), \varepsilon) \lesssim \mathrm{vc}(\mathcal{F}) \log \frac{2}{\varepsilon}.$$

When we substitute this into (8.30), we get the integral of $\sqrt{\log(2/\varepsilon)}$, which is bounded by an absolute constant. This gives

$$\frac{2}{\sqrt{n}}\,\mathbb{E}\sup_{f\in\mathcal{F}} Z_f \lesssim \sqrt{\frac{\mathrm{vc}(\mathcal{F})}{n}},$$

as required. □

Exercise 8.3.25 (Reinstating absolute value)**.** ☕☕☕ In the proof above, we bounded $\mathbb{E}\sup_{f\in\mathcal{F}} Z_f$ instead of $\mathbb{E}\sup_{f\in\mathcal{F}} |Z_f|$. Give a bound for the latter quantity.

Hint: Add the zero function to the class $\mathcal{F}$ and use Remark 8.1.5 to bound $|Z_f| = |Z_f - Z_0|$. Can the addition of one (zero) function significantly increase the VC dimension of $\mathcal{F}$?

Let us examine an important application of Theorem 8.3.23, which is called Glivenko-Cantelli Theorem. It addresses one of the most basic problems in statistics: how can we estimate the distribution of a random variable by sampling? Let X be a random variable with unknown cumulative distribution function (CDF)

$$F(x) = \mathbb{P}\left\{X \le x\right\}, \quad x \in \mathbb{R}.$$

Suppose we have a sample $X_1, \dots, X_n$ of i.i.d. random variables drawn from the same distribution as X. Then we can hope that $F(x)$ could be estimated by computing the fraction of the sample points satisfying $X_i \le x$, i.e. by the *empirical distribution function*

$$F_n(x) := \frac{\left|\{i \in [n] :\ X_i \le x\}\right|}{n}, \quad x \in \mathbb{R}.$$

Note that $F_n(x)$ is a *random* function.

[9] Recall that we have already encountered the *empirical measure* μ_n and the $L^2(\mu_n)$ norm a few times before, in particular in Lemma 8.3.19 and its proof, as well as in (8.23).

[10] The diameter of $\mathcal{F}$ gives the upper limit according to (8.13); check that the diameter is indeed bounded by 1.

The quantitative law of large numbers gives

$$\mathbb{E}\,|F_n(x) - F(x)| \le \frac{C}{\sqrt{n}} \quad \text{for every } x \in \mathbb{R}.$$

(Check this! Recall the variance computation in Section 1.3, but do it for the indicator random variables $\mathbf{1}_{\{X_i \le x\}}$ instead of X_i.)

Glivenko-Cantelli theorem is a stronger statement, which says that F_n approximates F *uniformly* over $x \in \mathbb{R}$.

Theorem 8.3.26 (Glivenko-Cantelli Theorem[11])**.** *Let $X_1, \ldots, X_n$ be independent random variables with common cumulative distribution function F. Then*

$$\mathbb{E}\,\|F_n - F\|_\infty = \mathbb{E} \sup_{x \in \mathbb{R}} |F_n(x) - F(x)| \le \frac{C}{\sqrt{n}}.$$

Proof This result is a particular case of Theorem 8.3.23. Indeed, let $\Omega = \mathbb{R}$, let $\mathcal{F}$ consist of the indicators of all half-bounded intervals, i.e.

$$\mathcal{F} := \left\{ \mathbf{1}_{(-\infty,x]} : \; x \in \mathbb{R} \right\},$$

and let the measure μ be the distribution[12] of X_i. As we know from Example 8.3.2, $\mathrm{vc}(\mathcal{F}) \le 2$. Thus Theorem 8.3.23 immediately implies the conclusion. □

Example 8.3.27 (Discrepancy)**.** Glivenko-Cantelli theorem can be easily generalized to random vectors. (Do it!) Let us give an illustration for $\mathbb{R}^2$. Draw a sample of i.i.d. points $X_1, \ldots, X_n$ from the uniform distribution on the unit square $[0,1]^2$ on the plane, see Figure 8.8. Consider the class $\mathcal{F}$ of indicators of all circles in that square. From Exercise 8.3.6 we know that $\mathrm{vc}(\mathcal{F}) = 3$. (Why does intersecting with the square does not affect the VC dimension?)

Apply Theorem 8.3.23. The sum $\sum_{i=1}^n f(X_i)$ is just the number of points in the circle with indicator function f, and the expectation $\mathbb{E}\, f(X)$ is the area of that circle. Then we can interpret the conclusion of Theorem 8.3.23 as follows. With high probability, a random sample of points $X_1, \ldots, X_n$ satisfies the following: for every circle $\mathcal{C}$ in the square $[0,1]^2$,

$$\text{number of points in } \mathcal{C} = \mathrm{Area}(\mathcal{C}) \cdot n + O(\sqrt{n}).$$

This is an example of a result in *geometric discrepancy* theory. The same result holds not only for circles but for half-planes, rectangles, squares, triangles, and polygons with $O(1)$ vertices, and any other class with bounded VC dimension.

[11] The classical statement of Glivenko-Cantelli theorem is about almost sure convergence, which we do not give here. However, it can be obtained from a high-probability version of the same argument using Borel-Cantelli lemma.

[12] Precisely, we define $\mu(A) := \mathbb{P}\left\{X \in A\right\}$ for every (Borel) subset $A \subset \mathbb{R}$.

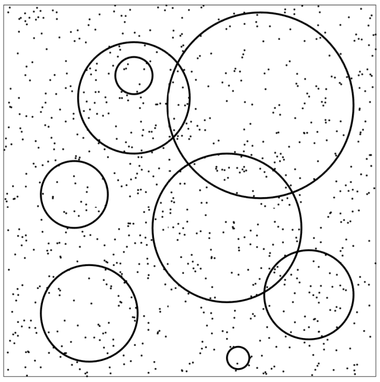


Figure 8.8 According to the uniform deviation inequality from Theorem 8.3.23, all circles have a fair share of the random sample of points. The number of points in each circle is proportional to its area with $O(\sqrt{n})$ error.

Remark 8.3.28 (Uniform Glivenko-Cantelli classes)**.** A class of real-valued functions $\mathcal{F}$ on a set Ω is called a *uniform Glivenko-Cantelli class* if, for any $\varepsilon > 0$,

$$\lim_{n\to\infty} \sup_{\mu} \mathbb{P} \left\{ \sup_{f\in\mathcal{F}} \left| \frac{1}{n} \sum_{i=1}^{n} f(X_i) - \mathbb{E}\, f(X) \right| > \varepsilon \right\} = 0,$$

where the supremum is taken over all probability measures μ on Ω and the points $X, X_1, \ldots, X_n$ are sampled from Ω according to the law μ. Theorem 8.3.23 followed by Markov's inequality yields the conclusion that any class of Boolean functions with finite VC dimension is uniform Glivenko-Cantelli.

Exercise 8.3.29 (Sharpness)**.** ☕☕☕ Prove that any class of Boolean functions with infinite VC dimension is not uniform Glivenko-Cantelli.

Hint: Choose a subset $\Lambda \subset \Omega$ of arbitrarily large cardinality d that is shattered by $\mathcal{F}$, and let μ be the uniform measure on Λ, assigning probability $1/d$ to each point.

Exercise 8.3.30 (A simpler, weaker bound)**.** ☕☕☕ Use Sauer-Shelah Lemma directly, instead of Pajor's Lemma, to prove a weaker version of the uniform deviation inequality (8.29) with

$$C\sqrt{\frac{d}{n}\log\frac{en}{d}}$$

in the right hand side, where $d = \mathrm{vc}(\mathcal{F})$.

Hint: Proceed similarly to the proof of Theorem 8.3.23. Combine a concentration inequality with a union bound over the entire class $\mathcal{F}$. Control the cardinality of $\mathcal{F}$ using Sauer-Shelah Lemma.

8.4 Application: statistical learning theory

Statistical learning theory, or machine learning, allows one to make predictions based on data. A typical problem of statistical learning can be stated mathematically as follows. Consider a function $T : \Omega \to \mathbb{R}$ on some set Ω, which we call a *target function*. Suppose T is unknown. We would like to learn T from its values on a finite sample of points $X_1, \ldots, X_n \in \Omega$. We assume that these points are independently sampled according to some common probability distribution $\mathbb{P}$ on Ω. Thus, our *training data* is

$$(X_i, T(X_i)), \quad i = 1, \ldots, n. \tag{8.31}$$

Our ultimate goal is to use the training data to make a good *prediction* of $T(X)$ for a new random point $X \in \Omega$, which was not in the training sample but is sampled from the same distribution; see Figure 8.9 for illustration.

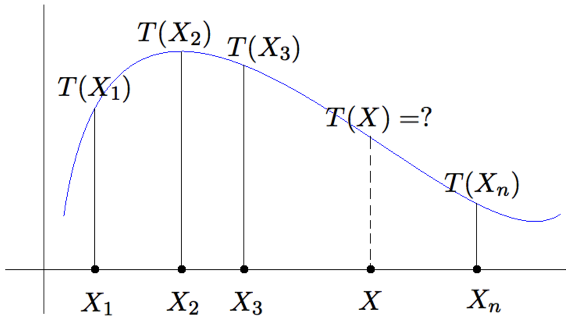


Figure 8.9 In a general learning problem, we are trying to learn an unknown function $T : \Omega \to \mathbb{R}$ (a "target function") from its values on a training sample $X_1, \ldots, X_n$ of i.i.d. points. The goal is to predict $T(X)$ for a new random point X.

You may notice some similarity between learning problems and Monte-Carlo integration, which we studied in Section 8.2.1. In both problems, we are trying to infer something about a function from its values on a random sample of points. But now our task is more difficult, as we are trying to learn the function itself and not just its integral, or average value, on Ω.

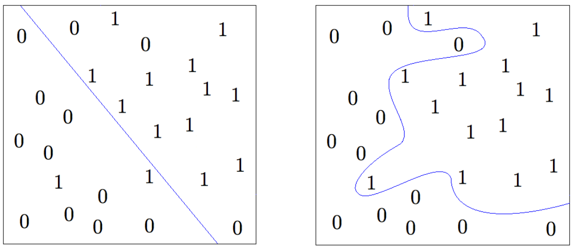
8.4.1 Classification problems

An important class of learning problems are classification problems, where the function T is Boolean (takes values 0 and 1), and thus T classifies points in Ω into two classes.

Example 8.4.1. Consider a health study on a sample of n patients. We record d various health parameters of each patient, such as blood pressure, body temperature, etc., arranging them as a vector $X_i \in \mathbb{R}^d$. Suppose we also know whether each of these patients has diabetes, and we encode this information as a binary number $T(X_i) \in \{0,1\}$ ($0 =$ healthy, $1 =$sick). Our goal is to learn from this training sample how to diagnose diabetes. We want to learn the target function $T : \mathbb{R}^d \to \{0,1\}$, which would output the diagnosis for *any* person based on his or her d health parameters.

For one more example, the vector X_i may contain the d gene expressions of i-th patient. Our goal is to learn is to diagnose a certain disease based on the patient's genetic information.

Figure 8.10c illustrates a classification problem where X is a random vector on the plane and the label Y can take values 0 and 1 like in Example 8.4.1. A solution of this classification problem can be described as a partition of the plane into two regions, one where $f(X) = 0$ (healthy) and another where $f(X) = 1$ (sick). Based on this solution, one can diagnose new patients by looking at which region their parameter vectors X fall in.



(a) Underfitting (b) Overfitting (c) Right fit

Figure 8.10 Trade-off between fit and complexity

8.4.2 Risk, fit and complexity

A solution to the learning problem can be expressed as a function $f : \Omega \to \mathbb{R}$. We would naturally want f to be as close to the target T as possible, so we would like to choose f that minimizes the *risk*

$$R(f) := \mathbb{E}\left(f(X) - T(X)\right)^2. \tag{8.32}$$

Here X denotes a random variable with distribution $\mathbb{P}$, i.e. with the same distribution as the sample points $X_1, \ldots, X_n \in \Omega$.

Example 8.4.2. In classification problems, T and f are Boolean functions, and thus

$$R(f) = \mathbb{P}\{f(X) \neq T(X)\}. \tag{8.33}$$

(Check!) So the risk is just the probability of misclassification, e.g. the misdiagnosis for a patient.

How much data do we need to learn, i.e. how large sample size n needs to be? This depends on the *complexity* of the problem. We need more data if we believe that the target function $T(X)$ may depend on X in an intricate way; otherwise we need less. Usually we do not know the complexity a priori. So we may restrict the complexity of the candidate functions f, insisting that our solution f must belong to some given class of functions $\mathcal{F}$ called the *hypothesis space*.

But how do we choose the hypothesis space $\mathcal{F}$ for a learning problem at hand? Although there is no general rule, the choice of $\mathcal{F}$ should be based on the *trade-off between fit and complexity*. Suppose we choose $\mathcal{F}$ to be too small; for example, we insist that the interface between healthy $(f(x) = 0)$ and sick diagnoses $(f(x) = 1)$ be a line, like in Figure 8.10a. Although we can learn such a simple function f with less data, we have probably oversimplified the problem. The linear functions do not capture the essential trends in this data, and this will lead to a big risk $R(f)$.

If, on the opposite, we choose $\mathcal{F}$ to be too large, this may result in *overfitting* where we essentially fit f to noise like in Figure 8.10b. Plus in this case we need a lot of data to learn such complicated functions.

A good choice for $\mathcal{F}$ is one that avoids either underfitting or overfitting, and captures the essential trends in the data just like in Figure 8.10c.

8.4.3 Empirical risk

What would be an optimal solution to the learning problem based on the training data? Ideally, we would like to find a function f^* from the hypothesis space $\mathcal{F}$ which would minimize the risk[13] $R(f) = \mathbb{E}\left(f(X) - T(X)\right)^2$, that is

$$f^* := \arg\min_{f \in \mathcal{F}} R(f).$$

If we are lucky and chose the hypothesis space $\mathcal{F}$ so that it contains the target function T, then risk is zero. Unfortunately, we can not compute the risk $R(f)$ and thus f^* from the training data. But we can try to *estimate* $R(f)$ and f^*.

Definition 8.4.3. The *empirical risk* for a function $f : \Omega \to \mathbb{R}$ is defined as

$$R_n(f) := \frac{1}{n} \sum_{i=1}^{n} \left(f(X_i) - T(X_i)\right)^2. \tag{8.34}$$

Denote by f_n^* a function in the hypothesis space $\mathcal{F}$ which minimizes the empirical risk:

$$f_n^* := \arg\min_{f \in \mathcal{F}} R_n(f),$$

Note that both $R_n(f)$ and f_n^* can be computed from the data. The outcome of learning from the data is thus f_n^*. The main question is: how large is the *excess risk*

$$R(f_n^*) - R(f^*)$$

[13] We assume for simplicity that the minimum is attained; an approximate minimizer could be used as well.

produced by our having to learn from a finite sample of size n? We give an answer in the next section.

8.4.4 Bounding the excess risk by the VC dimension

Let us specialize to the classification problems where the target T is a Boolean function.

Theorem 8.4.4 (Excess risk via VC dimension). *Assume that the target T is a Boolean function, and the hypothesis space $\mathcal{F}$ is a class of Boolean functions with finite VC dimension $\mathrm{vc}(\mathcal{F}) \geq 1$. Then*

$$\mathbb{E}\, R(f_n^*) \leq R(f^*) + C\sqrt{\frac{\mathrm{vc}(\mathcal{F})}{n}}.$$

We deduce this theorem from a uniform deviation inequality that we proved in Theorem 8.3.23. The following elementary observation will help us connect these two results.

Lemma 8.4.5 (Excess risk via uniform deviations). *We have*

$$R(f_n^*) - R(f^*) \leq 2 \sup_{f \in \mathcal{F}} |R_n(f) - R(f)|$$

pointwise.

Proof Denote $\varepsilon := \sup_{f \in \mathcal{F}} \big|R_n(f) - R(f)\big|$. Then

$$\begin{aligned} R(f_n^*) &\leq R_n(f_n^*) + \varepsilon \quad &&(\text{since } f_n^* \in \mathcal{F} \text{ by construction}) \\ &\leq R_n(f^*) + \varepsilon \quad &&(\text{since } f_n^* \text{ minimizes } R_n \text{ in the class } \mathcal{F}) \\ &\leq R(f^*) + 2\varepsilon \quad &&(\text{since } f^* \in \mathcal{F} \text{ by construction}). \end{aligned}$$

Subtracting $R(f^*)$ from both sides, we get the desired inequality. □

Proof of Theorem 8.4.4 By Lemma 8.4.5, it will be enough to show that

$$\mathbb{E} \sup_{f \in \mathcal{F}} |R_n(f) - R(f)| \lesssim \sqrt{\frac{\mathrm{vc}(\mathcal{F})}{n}}.$$

Recalling the definitions (8.34) and (8.32) of the empirical and true (population) risk, we express the left side as

$$\mathbb{E} \sup_{\ell \in \mathcal{L}} \left| \frac{1}{n} \sum_{i=1}^{n} \ell(X_i) - \mathbb{E}\, \ell(X) \right| \tag{8.35}$$

where $\mathcal{L}$ is the class of Boolean functions defined as

$$\mathcal{L} = \{(f - T)^2 : \; f \in \mathcal{F}\}.$$

The uniform deviation bound from Theorem 8.3.23 could be used at this point, but it only would give a bound in terms of the VC dimension of $\mathcal{L}$, which is not clear how to relate back to the VC dimension of $\mathcal{F}$. Instead, let us recall that in the proof Theorem 8.3.23, we first bounded (8.35) by

$$\frac{1}{\sqrt{n}}\,\mathbb{E}\int_0^1 \sqrt{\log \mathcal{N}(\mathcal{L}, L^2(\mu_n), \varepsilon)}\, d\varepsilon \tag{8.36}$$

up to an absolute constant factor. It is not hard to see that the covering numbers of $\mathcal{L}$ and $\mathcal{F}$ are related by the inequality

$$\mathcal{N}(\mathcal{L}, L^2(\mu_n), \varepsilon) \le \mathcal{N}(\mathcal{F}, L^2(\mu_n), \varepsilon) \quad \text{for any } \varepsilon \in (0,1). \tag{8.37}$$

(We check this inequality accurately in Exercise 8.4.6.) So we may replace $\mathcal{L}$ by $\mathcal{F}$ in (8.36) paying the price of an absolute constant factor (check!). We then follow the rest of the proof of Theorem 8.3.23 and conclude that (8.36) is bounded by

$$\sqrt{\frac{\mathrm{vc}(\mathcal{F})}{n}}$$

as we desired. □

Exercise 8.4.6. ☕☕ Check the inequality (8.37).

Hint: Check that for any triple of Boolean functions f, g, T satisfies the identity $\big((f-T)^2 - (g-T)^2\big)^2 = (f-g)^2$.

8.4.5 Interpretation and examples

What does Theorem 8.4.4 really say about learning? It quantifies the risk of having to learn from limited data, which we called excess risk. Theorem 8.4.4 states that on average, the excess risk of learning from a finite sample of size n is proportional to $\sqrt{\mathrm{vc}(\mathcal{F})/n}$. Equivalently, if we want to bound the expected excess risk by ε, all we need to do is take a sample of size

$$n \asymp \varepsilon^{-2}\,\mathrm{vc}(\mathcal{F}).$$

This result answers the question of how much training data we need for learning. And the answer is: *it is enough to have the sample size n exceed the VC dimension of the hypothesis class* $\mathcal{F}$ (up to some constant factor).

Let us illustrate this principle by thoroughly working out a specific learning problem from Figure 8.10. We are trying to learn an unknown function $T : \mathbb{R}^2 \to \{0,1\}$. This is a classification problem, where the function T assigns labels 0 and 1 to the points on the plane, and we are trying to learn those labels.

First, we collect training data – some n points $X_1, \ldots, X_n$ on the plane whose labels $T(X_i)$ we know. We assume that the points X_i are sampled at random according to some probability distribution $\mathbb{P}$ on the plane.

Next, we need to choose a hypothesis space $\mathcal{F}$. This is a class of Boolean functions on the plane where we will be looking for a solution to our learning problem. We need to make sure that $\mathcal{F}$ is neither too large (to prevent overfitting) nor too small (to prevent underfitting). We may expect that the interface between the two classes is a nontrivial but not too intricate curve, such as an arc in

Figure 8.10c. For example, it may be reasonable to include in $\mathcal{F}$ the indicator functions of all circles on the plane.[14] So let us choose

$$\mathcal{F} := \{\mathbf{1}_C : \text{ circles } C \subset \mathbb{R}^2\}. \tag{8.38}$$

Recall from Exercise 8.3.6 that $\mathrm{vc}(\mathcal{F}) = 3$.

Next, we set up the empirical risk

$$R_n(f) := \frac{1}{n}\sum_{i=1}^{n}\left(f(X_i) - T(X_i)\right)^2.$$

We can compute the empirical risk from data for any given function f on the plane. Finally, we minimize the empirical risk over our hypothesis class $\mathcal{F}$, and thus compute

$$f_n^* := \arg\min_{f\in\mathcal{F}} R_n(f).$$

Exercise 8.4.7. ☕☕ Check that f_n^* is a function in $\mathcal{F}$ that minimizes the number of data points X_i where the function disagrees with the labels $T(X_i)$.

We output the function f_n^* as the solution of the learning problem. By computing $f_n^*(x)$, we can make a prediction for the label of the points x that were not in the training set.

How reliable is this prediction? We quantified the predicting power of a Boolean function f with the concept of *risk* $R(f)$. It gives the probability that f assigns the wrong label to a random point X sampled from the same distribution on the plane as the data points:

$$R(f) = \mathbb{P}\{f(X) \neq T(X)\}.$$

Using Theorem 8.4.4 and recalling that $\mathrm{vc}(\mathcal{F}) = 3$, we get a bound on the risk for our solution f_n^*:

$$\mathbb{E}\, R(f_n^*) \le R(f^*) + \frac{C}{\sqrt{n}}.$$

Thus, on average, our solution f_n^* gives correct predictions almost with the same probability – within $1/\sqrt{n}$ error – as the best available function f^* in the hypothesis class $\mathcal{F}$, i.e. the best chosen circle.

Exercise 8.4.8 (Random outputs)**.** ☕☕☕ Our model of a learning problem (8.31) postulates that the output $T(X)$ must be completely determined by the input X. This is rarely the case in practice. For example, it is not realistic to assume that the diagnosis $T(X) \in \{0, 1\}$ of a disease is completely determined by the available genetic information X. What is more often true is that the output Y is a random variable, which is correlated with the input X; the goal of learning is still to predict Y from X as best as possible.

[14] We can also include all half-spaces, which we can think of circles with infinite radii centered at infinity.

Extend the theory of learning leading up to Theorem 8.4.4 for the training data of the form

$$(X_i, Y_i), \quad i = 1, \dots, n$$

where (X_i, Y_i) are independent copies of a pair (X, Y) consisting of an input random point $X \in \Omega$ and an output random variable Y.

Exercise 8.4.9 (Learning in the class of Lipschitz functions). Consider the hypothesis class of Lipschitz functions

$$\mathcal{F} := \{f : [0,1] \to [0,1], \; \|f\|_{\mathrm{Lip}} \le L\}$$

and a target function $T : [0,1] \to [0,1]$.

(a) Show that the random process $X_f := R_n(f) - R(f)$ has sub-gaussian increments:

$$\|X_f - X_g\|_{\psi_2} \le \frac{C(L)}{\sqrt{n}} \|f - g\|_\infty \quad \text{for all } f, g \in \mathcal{F}.$$

(b) Use Dudley's inequality to deduce that

$$\mathbb{E} \sup_{f \in \mathcal{F}} \left| R_n(f) - R(f) \right| \le \frac{C(L)}{\sqrt{n}}.$$

Hint: Proceed like in the proof of Theorem 8.2.3.

(c) Conclude that the excess risk satisfies

$$\mathbb{E}\, R(f_n^*) - R(f^*) \le \frac{C(L)}{\sqrt{n}}.$$

The value of $C(L)$ may be different in different parts of the exercise, but it may only depend on L.

8.5 Generic chaining

Dudley's inequality is a simple and useful tool for bounding a general random process. Unfortunately, as we saw in Exercise 8.1.12, Dudley's inequality can be loose. The reason behind this is that the covering numbers $\mathcal{N}(T, d, \varepsilon)$ do not contain enough information to control the magnitude of $\mathbb{E} \sup_{t \in T} X_t$.

8.5.1 A makeover of Dudley's inequality

Fortunately, there *is* a way to obtain accurate, two-sided bounds on $\mathbb{E} \sup_{t \in T} X_t$ for sub-gaussian processes $(X_t)_{t \in T}$ in terms of the geometry of T. This method is called *generic chaining*, and it is essentially a sharpening of the chaining method we developed in the proof of Dudley's inequality (Theorem 8.1.4). Recall that the outcome of chaining was the bound (8.12):

$$\mathbb{E} \sup_{t \in T} X_t \lesssim \sum_{k=\kappa+1}^{\infty} \varepsilon_{k-1} \sqrt{\log |T_k|}. \tag{8.39}$$

Here ε_k are decreasing positive numbers and T_k are ε_k-nets of T such that $|T_\kappa| = 1$. To be specific, in the proof of Theorem 8.1.4 we chose

$$\varepsilon_k = 2^{-k} \quad \text{and} \quad |T_k| = \mathcal{N}(T, d, \varepsilon_k),$$

so $T_k \subset T$ were the smallest ε_k-nets of T.

In preparation for generic chaining, let us now turn around our choice of ε_k and T_k. Instead of fixing ε_k and operating with the smallest possible cardinality of T_k, let us fix the cardinality of T_k and operate with the smallest possible ε_k. Namely, let us fix some subsets $T_k \subset T$ such that

$$|T_0| = 1, \quad |T_k| \le 2^{2^k}, \quad k = 1, 2, \dots \tag{8.40}$$

Such sequence of sets $(T_k)_{k=0}^\infty$ is called an *admissible sequence*. Put

$$\varepsilon_k = \sup_{t \in T} d(t, T_k),$$

where $d(t, T_k)$ denotes the distance[15] from t to the set T_k. Then each T_k is an ε_k-net of T. With this choice of ε_k and T_k, the chaining bound (8.39) becomes

$$\mathbb{E} \sup_{t \in T} X_t \lesssim \sum_{k=1}^\infty 2^{k/2} \sup_{t \in T} d(t, T_{k-1}).$$

After re-indexing, we conclude

$$\mathbb{E} \sup_{t \in T} X_t \lesssim \sum_{k=0}^\infty 2^{k/2} \sup_{t \in T} d(t, T_k). \tag{8.41}$$

8.5.2 Talagrand's γ_2 functional and generic chaining

So far, nothing has really happened. The bound (8.41) is just an equivalent way to state Dudley's inequality. The important step will come now. The generic chaining will allow us to pull the supremum *outside* the sum in (8.41). The resulting important quantity has a name:

Definition 8.5.1 (Talagrand's γ_2 functional)**.** Let (T, d) be a metric space. A sequence of subsets $(T_k)_{k=0}^\infty$ of T is called an *admissible sequence* if the cardinalities of T_k satisfy (8.40). The γ_2 *functional* of T is defined as

$$\gamma_2(T, d) = \inf_{(T_k)} \sup_{t \in T} \sum_{k=0}^\infty 2^{k/2} d(t, T_k)$$

where the infimum is with respect to all admissible sequences.

Since the supremum in the γ_2 functional is outside the sum, it is smaller than the Dudley's sum in (8.41). The difference between the γ_2 functional and the Dudley's sum can look minor, but sometimes it is real:

[15] Formally, the distance from a point $t \in T$ to a subset $A \subset T$ in a metric space T is defined as $d(t, A) := \inf\{d(t, a) : a \in A\}$.

Exercise 8.5.2 (γ_2 functional and Dudley's sum)**.** ☕☕☕ Consider the same set $T \subset \mathbb{R}^n$ as in Exercise 8.1.12, i.e.

$$T := \{0\} \cup \left\{ \frac{e_k}{\sqrt{1+\log k}},\ k = 1, \dots, n \right\}.$$

(a) Show that the γ_2-functional of T (with respect to the Euclidean metric) is bounded, i.e.

$$\gamma_2(T,d) = \inf_{(T_k)} \sup_{t \in T} \sum_{k=0}^{\infty} 2^{k/2} d(t, T_k) \le C.$$

Hint: Use the first 2^{2^k} vectors in T to define T_k.

(b) Check that Dudley's sum is unbounded, i.e.

$$\inf_{(T_k)} \sum_{k=0}^{\infty} 2^{k/2} \sup_{t \in T} d(t, T_k) \to \infty$$

as $n \to \infty$.

We now state an improvement of Dudley's inequality, in which Dudley's sum (or integral) is replaced by a tighter quantity, the γ_2-functional.

Theorem 8.5.3 (Generic chaining bound)**.** *Let $(X_t)_{t \in T}$ be a mean zero random process on a metric space (T, d) with sub-gaussian increments as in (8.1). Then*

$$\mathbb{E} \sup_{t \in T} X_t \le CK \gamma_2(T, d).$$

Proof We proceed with the same chaining method that we introduced in the proof of Dudley's inequality Theorem 8.1.4, but we will do chaining more accurately.

Step 1: Chaining set-up. As before, we may assume that $K = 1$ and that T is finite. Let (T_k) be an admissible sequence of subsets of T, and denote $T_0 = \{t_0\}$. We walk from t_0 to a general point $t \in T$ along the chain

$$t_0 = \pi_0(t) \to \pi_1(t) \to \cdots \to \pi_K(t) = t$$

of points $\pi_k(t) \in T_k$ that are chosen as best approximations to t in T_k, i.e.

$$d(t, \pi_k(t)) = d(t, T_k).$$

The displacement $X_t - X_{t_0}$ can be expressed as a telescoping sum similar to (8.9):

$$X_t - X_{t_0} = \sum_{k=1}^{K} (X_{\pi_k(t)} - X_{\pi_{k-1}(t)}). \tag{8.42}$$

Step 2: Controlling the increments. This is where we need to be more accurate than in Dudley's inequality. We would like to have a uniform bound on the increments, a bound that would state with high probability that

$$\left| X_{\pi_k(t)} - X_{\pi_{k-1}(t)} \right| \le 2^{k/2} d(t, T_k) \quad \forall k \in \mathcal{N}, \quad \forall t \in T. \tag{8.43}$$

Summing these inequalities over all k would lead to a desired bound in terms of $\gamma_2(T,d)$.

To prove (8.43), let us fix k and t first. The sub-gaussian assumption tells us that

$$\left\|X_{\pi_k(t)} - X_{\pi_{k-1}(t)}\right\|_{\psi_2} \le d(\pi_k(t), \pi_{k-1}(t)).$$

This means that for every $u \ge 0$, the event

$$\left|X_{\pi_k(t)} - X_{\pi_{k-1}(t)}\right| \le Cu2^{k/2} d(\pi_k(t), \pi_{k-1}(t)) \tag{8.44}$$

holds with probability at least

$$1 - 2\exp(-8u^2 2^k).$$

(To get the constant 8, choose the absolute constant C large enough.)

We can now unfix $t \in T$ by taking a union bound over

$$|T_k| \cdot |T_{k-1}| \le |T_k|^2 \le 2^{2^{k+1}}$$

possible pairs $(\pi_k(t), \pi_{k-1}(t))$. Similarly, we can unfix k by a union bound over all $k \in \mathbb{N}$. Then the probability that the bound (8.44) holds simultaneously for all $t \in T$ and $k \in \mathbb{N}$ is at least

$$1 - \sum_{k=1}^{\infty} 2^{2^{k+1}} \cdot 2\exp(-8u^2 2^k) \ge 1 - 2\exp(-u^2).$$

if $u > c$. (Check the last inequality!)

Step 3: Summing up the increments. In the event that the bound (8.44) does holds for all $t \in T$ and $k \in \mathbb{N}$, we can sum up the inequalities over $k \in \mathcal{N}$ and plug the result into the chaining sum (8.42). This yields

$$|X_t - X_{t_0}| \le Cu \sum_{k=1}^{\infty} 2^{k/2} d(\pi_k(t), \pi_{k-1}(t)). \tag{8.45}$$

By triangle inequality, we have

$$d(\pi_k(t), \pi_{k-1}(t)) \le d(t, \pi_k(t)) + d(t, \pi_{k-1}(t)).$$

Using this bound and doing re-indexing, we find that the right hand side of (8.45) can be bounded by $\gamma_2(T,d)$, that is

$$|X_t - X_{t_0}| \le C_1 u \gamma_2(T,d).$$

(Check!) Taking the supremum over T yields

$$\sup_{t \in T} |X_t - X_{t_0}| \le C_2 u \gamma_2(T,d).$$

Recall that this inequality holds with probability at least $1 - 2\exp(-u^2)$ for any $u > c$. This means that the magnitude in question is a sub-gaussian random variable:

$$\left\| \sup_{t \in T} |X_t - X_{t_0}| \right\|_{\psi_2} \le C_3 \gamma_2(T,d).$$

This quickly implies the conclusion of Theorem 8.5.3. (Check!) □

Remark 8.5.4 (Supremum of increments)**.** Similarly to Dudley's inequality (Remark 8.1.5), the generic chaining also gives the uniform bound

$$\mathbb{E} \sup_{t,s\in T} |X_t - X_s| \le CK\gamma_2(T, d).$$

which is valid even without the mean zero assumption $\mathbb{E} X_t = 0$.

The argument above gives not only a bound on expectation but also a tail bound for $\sup_{t\in T} X_t$. Let us now give a better tail bound, similar to the one we had in Theorem 8.1.6 for Dudley's inequality.

Theorem 8.5.5 (Generic chaining: tail bound)**.** *Let $(X_t)_{t\in T}$ be a random process on a metric space (T, d) with sub-gaussian increments as in (8.1). Then, for every $u \ge 0$, the event*

$$\sup_{t,s\in T} |X_t - X_s| \le CK\Big[\gamma_2(T, d) + u \cdot \operatorname{diam}(T)\Big]$$

holds with probability at least $1 - 2\exp(-u^2)$.

Exercise 8.5.6. ☕☕☕☕ Prove Theorem 8.5.5. To this end, state and use a variant of the increment bound (8.44) with $u + 2^{k/2}$ instead of $u2^{k/2}$. In the end of the argument, you will need a bound on the sum of steps $\sum_{k=1}^{\infty} d(\pi_k(t), \pi_{k-1}(t))$. For this, modify the chain $\{\pi_k(t)\}$ by doing a "lazy walk" on it. Stay at the current point $\pi_k(t)$ for a few steps (say, $q-1$) until the distance to t improves by a factor of 2, that is until

$$d(t, \pi_{k+q}(t)) \le \frac{1}{2} d(t, \pi_k(t)),$$

then jump to $\pi_{k+q}(t)$. This will make the sum of the steps geometrically convergent.

Exercise 8.5.7 (Dudley's integral vs. γ_2 functional)**.** ☕☕☕ Show that γ_2 functional is bounded by Dudley's integral. Namely, show that for any metric space (T, d), one has

$$\gamma_2(T, d) \le C \int_0^{\infty} \sqrt{\log \mathcal{N}(T, d, \varepsilon)}\, d\varepsilon.$$

8.6 Talagrand's majorizing measure and comparison theorems

Talagrand's γ_2 functional introduced in Definition 8.5.1 has some advantages and disadvantages over Dudley's integral. A disadvantage is that $\gamma_2(T, d)$ is usually harder to compute than the metric entropy that defines Dudley's integral. Indeed, it could take a real effort to construct a good admissible sequence of sets. However, unlike Dudley's integral, the γ_2 functional gives a bound on Gaussian processes that is *optimal* up to an absolute constant. This is the content of the following theorem.

Theorem 8.6.1 (Talagrand's majorizing measure theorem). *Let $(X_t)_{t\in T}$ be a mean zero Gaussian process on a set T. Consider the canonical metric defined on T by (7.13), i.e. $d(t,s) = \|X_t - X_s\|_{L^2}$. Then*

$$c \cdot \gamma_2(T,d) \le \mathbb{E} \sup_{t\in T} X_t \le C \cdot \gamma_2(T,d).$$

The upper bound in Theorem 8.6.1 follows directly from generic chaining (Theorem 8.5.3). The lower bound is harder to obtain. Its proof, which we do not present in this book, can be thought of as a far reaching, multi-scale strengthening of Sudakov's inequality (Theorem 7.4.1).

Note that the upper bound, as we know from Theorem 8.5.3, holds for any *sub-gaussian* process. Therefore, by combining the upper and lower bounds together, we can deduce that any sub-gaussian process is bounded (via γ_2 functional) by a Gaussian process. Let us state this important comparison result.

Corollary 8.6.2 (Talagrand's comparison inequality). *Let $(X_t)_{t\in T}$ be a mean zero random process on a set T and let $(Y_t)_{t\in T}$ be a mean zero Gaussian process. Assume that for all $t,s \in T$, we have*

$$\|X_t - X_s\|_{\psi_2} \le K\|Y_t - Y_s\|_{L^2}.$$

Then

$$\mathbb{E} \sup_{t\in T} X_t \le CK\, \mathbb{E} \sup_{t\in T} Y_t.$$

Proof Consider the canonical metric on T given by $d(t,s) = \|Y_t - Y_s\|_{L^2}$. Apply the generic chaining bound (Theorem 8.5.3) followed by the lower bound in the majorizing measure Theorem 8.6.1. Thus we get

$$\mathbb{E} \sup_{t\in T} X_t \le CK\gamma_2(T,d) \le CK\, \mathbb{E} \sup_{t\in T} Y_t.$$

The proof is complete. □

Corollary 8.6.2 extends Sudakov-Fernique's inequality (Theorem 7.2.11) for sub-gaussian processes. All we pay for such extension is an absolute constant factor.

Let us apply Corollary 8.6.2 for a canonical Gaussian process

$$Y_x = \langle g, x\rangle, \quad x \in T,$$

defined on a subset $T \subset \mathbb{R}^n$. Recall from Section 7.5 that the magnitude of this process,

$$w(T) = \mathbb{E} \sup_{x\in T} \langle g, x\rangle$$

is the *Gaussian width* of T. We immediately obtain the following corollary.

Corollary 8.6.3 (Talagrand's comparison inequality: geometric form). *Let $(X_x)_{x\in T}$ be a mean zero random process on a subset $T \subset \mathbb{R}^n$. Assume that for all $x,y \in T$, we have*

$$\|X_x - X_y\|_{\psi_2} \le K\|x - y\|_2.$$

Then

$$\mathbb{E} \sup_{x \in T} X_x \le CKw(T).$$

Exercise 8.6.4 (Bound on absolute values). ☕☕ Let $(X_x)_{x\in T}$ be a random process (not necessarily mean zero) on a subset $T \subset \mathbb{R}^n$. Assume that[16] $X_0 = 0$, and for all $x, y \in T \cup \{0\}$ we have

$$\|X_x - X_y\|_{\psi_2} \le K\|x - y\|_2.$$

Prove that[17]

$$\mathbb{E} \sup_{x \in T} |X_x| \le CK\gamma(T).$$

Hint: Use Remark 8.5.4 and the majorizing measure theorem to get a bound in terms of the Gaussian width $w(T \cup \{0\})$, then pass to Gaussian complexity using Exercise 7.6.9.

Exercise 8.6.5 (Tail bound). ☕☕ Show that, in the setting of Exercise 8.6.4, for every $u \ge 0$ we have[18]

$$\sup_{x \in T} |X_x| \le CK\Big(w(T) + u \cdot \mathrm{rad}(T)\Big)$$

with probability at least $1 - 2\exp(-u^2)$.

Hint: Argue like in Exercise 8.6.4. Use Theorem 8.5.5 and Exercise 7.6.9.

Exercise 8.6.6 (Higher moments of the deviation). ☕ Check that, in the setting of Exercise 8.6.4,

$$\Big(\mathbb{E} \sup_{x \in T} |X_x|^p\Big)^{1/p} \le C\sqrt{p}\, K\gamma(T).$$

8.7 Chevet's inequality

Talagrand's comparison inequality (Corollary 8.6.2) has several important consequences. We cover one application now, others will appear later in this book.

In this section we look for a uniform bound for a random quadratic form, i.e. a bound on the quantity

$$\sup_{x \in T,\, y \in S} \langle Ax, y\rangle \tag{8.46}$$

where A is a random matrix and T and S are general sets.

We already encountered problems of this type when we analyzed the norms of random matrices, namely in the proofs of Theorems 4.4.5 and 7.3.1. In those situations, the sets T and S were Euclidean balls. This time, we let T and S be arbitrary geometric sets. Our bound on (8.46) will depend on just two geometric parameters of T and S: the *Gaussian width* and the *radius*, which we define as

$$\mathrm{rad}(T) := \sup_{x \in T} \|x\|_2. \tag{8.47}$$

[16] If the set T does not contain the origin, simply define $X_0 := 0$.
[17] Recall from Section 7.6.2 that $\gamma(T)$ is the Gaussian complexity of T.
[18] Here as usual $\mathrm{rad}(T)$ denotes the radius of T.

Theorem 8.7.1 (Sub-gaussian Chevet's inequality)**.** *Let A be an $m \times n$ random matrix whose entries A_{ij} are independent, mean zero, sub-gaussian random variables. Let $T \subset \mathbb{R}^n$ and $S \subset \mathbb{R}^m$ be arbitrary bounded sets. Then*

$$\mathbb{E} \sup_{x \in T,\, y \in S} \langle Ax, y \rangle \le CK \left[w(T) \operatorname{rad}(S) + w(S) \operatorname{rad}(T) \right]$$

where $K = \max_{ij} \|A_{ij}\|_{\psi_2}$.

Before we prove this theorem, let us make one simple illustration of its use. Setting $T = S^{n-1}$ and $S = S^{m-1}$, we recover a bound on the operator norm of A,

$$\mathbb{E}\, \|A\| \le CK(\sqrt{n} + \sqrt{m}),$$

which we obtained in Section 4.4.2 using a different method.

Proof of Theorem 8.7.1 We use the same method as in our proof of the sharp bound on Gaussian random matrices (Theorem 7.3.1). That argument was based on Sudakov-Fernique comparison inequality; this time, we use the more general Talagrand's comparison inequality.

Without loss of generality, $K = 1$. We would like to bound the random process

$$X_{uv} := \langle Au, v \rangle \,, \quad u \in T,\ v \in S.$$

Let us first show that this process has sub-gaussian increments. For any $(u, v), (w, z) \in T \times S$, we have

$$\begin{aligned}
\|X_{uv} - X_{wz}\|_{\psi_2} &= \Big\| \sum_{i,j} A_{ij}(u_i v_j - w_i z_j) \Big\|_{\psi_2} \\
&\le \Big(\sum_{i,j} \|A_{ij}(u_i v_j - w_i z_j)\|_{\psi_2}^2 \Big)^{1/2} \quad \text{(by Proposition 2.6.1)} \\
&\le \Big(\sum_{i,j} \|u_i v_j - w_i z_j\|_2^2 \Big)^{1/2} \quad \text{(since } \|A_{ij}\|_{\psi_2} \le K = 1) \\
&= \|uv^\mathsf{T} - wz^\mathsf{T}\|_F \\
&= \|(uv^\mathsf{T} - wv^\mathsf{T}) + (wv^\mathsf{T} - wz^\mathsf{T})\|_F \quad \text{(adding, subtracting)} \\
&\le \|(u-w)v^\mathsf{T}\|_F + \|w(v-z)^\mathsf{T}\|_F \quad \text{(by triangle inequality)} \\
&= \|u-w\|_2 \|v\|_2 + \|v-z\|_2 \|w\|_2 \\
&\le \|u-w\|_2 \operatorname{rad}(S) + \|v-z\|_2 \operatorname{rad}(T).
\end{aligned}$$

To apply Talagrand's comparison inequality, we need to choose a Gaussian process (Y_{uv}) to compare the process (X_{uv}) to. The outcome of our calculation of the increments of (X_{uv}) suggests the following definition for (Y_{uv}):

$$Y_{uv} := \langle g, u \rangle \operatorname{rad}(S) + \langle h, v \rangle \operatorname{rad}(T),$$

where

$$g \sim N(0, I_n), \quad h \sim N(0, I_m)$$

are independent Gaussian vectors. The increments of this process are

$$\|Y_{uv} - Y_{wz}\|_{L^2}^2 = \|u - w\|_2^2 \operatorname{rad}(S)^2 + \|v - z\|_2^2 \operatorname{rad}(T)^2.$$

(Check this as in the proof of Theorem 7.3.1.)

Comparing the increments of the two processes, we find that

$$\|X_{uv} - X_{wz}\|_{\psi_2} \lesssim \|Y_{uv} - Y_{wz}\|_2.$$

(Check!) Applying Talagrand's comparison inequality (Corollary 8.6.3), we conclude that

$$\begin{aligned}\mathbb{E} \sup_{u \in T,\, v \in S} X_{uv} &\lesssim \mathbb{E} \sup_{u \in T,\, v \in S} Y_{uv} \\ &= \mathbb{E} \sup_{u \in T} \langle g, u \rangle \operatorname{rad}(S) + \mathbb{E} \sup_{v \in S} \langle h, v \rangle \operatorname{rad}(T) \\ &= w(T) \operatorname{rad}(S) + w(S) \operatorname{rad}(T),\end{aligned}$$

as claimed. □

Chevet's inequality is optimal, up to an absolute constant factor.

Exercise 8.7.2 (Sharpness of Chevet's inequality)**.** ☕☕ Let A be an $m \times n$ random matrix whose entries A_{ij} are independent $N(0,1)$ random variables. Let $T \subset \mathbb{R}^n$ and $S \subset \mathbb{R}^m$ be arbitrary bounded sets. Show that the reverse of Chevet's inequality holds:

$$\mathbb{E} \sup_{x \in T,\, y \in S} \langle Ax, y \rangle \geq c \left[w(T) \operatorname{rad}(S) + w(S) \operatorname{rad}(T) \right].$$

Hint: Note that $\mathbb{E} \sup_{x \in T,\, y \in S} \langle Ax, y \rangle \geq \sup_{x \in T} \mathbb{E} \sup_{y \in S} \langle Ax, y \rangle$.

Exercise 8.7.3 (High probability version of Chevet)**.** ☕☕ Under the assumptions of Theorem 8.7.1, prove a tail bound for $\sup_{x \in T,\, y \in S} \langle Ax, y \rangle$.

Hint: Use the result of Exercise 8.6.5.

Exercise 8.7.4 (Gaussian Chevet's inequality)**.** ☕☕ Suppose the entries of A are $N(0,1)$. Show that Theorem 8.7.1 holds with sharp constant 1, that is

$$\mathbb{E} \sup_{x \in T,\, y \in S} \langle Ax, y \rangle \leq w(T) \operatorname{rad}(S) + w(S) \operatorname{rad}(T).$$

Hint: Use Sudakov-Fernique's inequality (Theorem 7.2.11) instead of Talagrand's comparison inequality.

8.8 Notes

The idea of chaining already appears in Kolmogorov's proof of his continuity theorem for Brownian motion, see e.g. [156, Chapter 1]. Dudley's integral inequality (Theorem 8.1.3) can be traced to the work of R. Dudley. Our exposition in Section 8.1 mostly follows [130, Chapter 11], [199, Section 1.2] and [212, Section 5.3]. The upper bound in Theorem 8.1.13 (a reverse Sudakov's inequality) seems to be a folklore result.

Monte-Carlo methods mentioned in Section 8.2 are extremely popular in scientific computing, especially when combined with the power of Markov chains, see e.g. [38]. In the same section we introduced the concept of empirical processes. The rich theory of empirical processes has applications to statistics and machine learning, see [211, 210, 171, 143]. In the terminology of empirical processes, Theorem 8.2.3) yields that the class of Lipschitz functions $\mathcal{F}$ is *uniform Glivenko-Cantelli*. Our presentation of this result (as well as relation to Wasserstein's distance and transportation) is loosely based on [212, Example 5.15]. For a deep introduction to transportation of measures, see [224].

The concept of VC dimension that we studied in Section 8.3 goes back to the foundational work of V. Vapnik and A. Chervonenkis [217]; modern treatments can be found e.g. in [211, Section 2.6.1], [130, Section 14.3], [212, Section 7.2], [138, Sections 10.2–10.3], [143, Section 2.2], [211, Section 2.6]. Pajor's Lemma 8.3.13 is originally due to A. Pajor [164]; see [79], [130, Proposition], [212, Theorem 7.19], [211, Lemma 2.6.2].

What we now call Sauer-Shelah Lemma (Theorem 8.3.16) was proved independently by V. Vapnik and A. Chervonenkis [217], N. Sauer [180] and M. Perles and S. Shelah [184]. Various proofs of Sauer-Shelah lemma can be found in literature, e.g. [25, Chapter 17], [138, Sections 10.2–10.3], [130, Section 14.3]. A number of variants of Sauer-Shelah Lemma is known, see e.g. [101, 196, 197, 6, 219].

Theorem 8.3.18 is due to R. Dudley [70]; see [130, Section 14.3], [211, Theorem 2.6.4]. The dimension reduction Lemma 8.3.19 is implicit in Dudley's proof; it was stated explicitly in [146] and reproduced in [212, Lemma 7.17]. for generalization of VC theory from $\{0, 1\}$ to general real-valued function classes, see [146, 176], [212, Sections 7.3–7.4].

Since the foundational work of V. Vapnik and A. Chervonenkis [217], bounds on empirical processes via VC dimension like Theorem 8.3.23 were in the spotlight of the statistical learning theory, see e.g. [143, 18, 211, 176], [212, Chapter 7]. Our presentation of Theorem 8.3.23 is based on [212, Corollary 7.18]. Although explicit statement of this result are difficult to find in earlier literature, it can be derived from [18, Theorem 6], [33, Section 5].

Glivenko-Cantelli theorem (Theorem 8.3.26) is a result from 1933 [82, 48] which predated and partly motivated the later development of VC theory; see [130, Section 14.2], [211, 71] for more on Glivenko-Cantelli theorems and other uniform results in probability theory. Example 8.3.27 discusses a basic problem in discrepancy theory; see [137] for a comprehensive treatment of discrepancy theory.

In Section 8.4 we scratch the surface of statistical learning theory, which is a big area on the intersection of probability, statistics, and theoretical computer science. For deeper introduction to this subject, see e.g. the tutorials [30, 143] and books [106, 99, 123].

Generic chaining, which we presented in Section 8.5 , was put forward by M. Talagrand since 1985 (after an earlier work of X. Fernique [75]) as a sharp method to obtain bounds on Gaussian processes. Our presentation is based on the book [199], which discusses ramifications, applications and history of generic chaining in great detail. The upper bound on sub-gaussian processes (Theorem 8.5.3) can be found in [199, Theorem 2.2.22]; the lower bound (the majorizing measure Theorem 8.6.1) can be found in [199, Theorem 2.4.1]. Talagrand's comparison inequality (Corollary 8.6.2) is borrowed from [199, Theorem 2.4.12]. Another presentation of generic chaining can be found in [212, Chapter 6]. A different proof of the majorizing measure theorem was recently given by R. van Handel in [214, 215]. A high-probability version of generic chaining bound (Theorem 8.5.5) is from [199, Theorem 2.2.27]; it was also proved by a different method by S. Dirksen [63].

In Section 8.7 we presented Chevet's inequality for sub-gaussian processes. In the existing literature, this inequality is stated only for Gaussian processes. It goes back to S. Chevet [54]; the constants were then improved by Y. Gordon [84], leading to the result we stated in Exercise 8.7.4. A exposition of this result can be found in [11, Section 9.4]. For variants and applications of Chevet's inequality, see [205, 2].